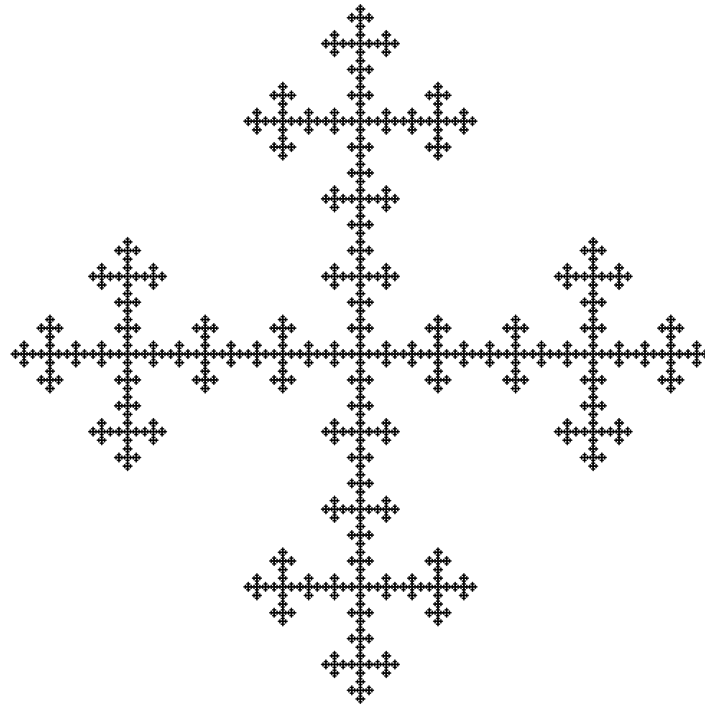


FRACTIONAL GAUSSIAN FIELDS ON THE VICSEK FRACTAL



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Abstract

In this thesis we will prove the convergence in distribution of discrete fractional Gaussian fields to fractional Gaussian fields on the Vicsek fractal for a space of suitable test functions.

We will introduce the notion of self-similar sets and their properties so as to construct the Vicsek fractal. We will also examine the concepts of Hausdorff, walk and spectral dimension, followed by a normalized Hausdorff measure for the Vicsek fractal.

Subsequently we will construct the discrete Laplacian, the Kigami Laplacian and the Dirichlet Laplacian, followed by the introduction of the discrete Riesz kernel and the fractional Riesz kernel. This will allow us to define the discrete fractional Laplacian.

Further notions of functional analysis are needed to construct and define semi-groups and heat kernels for densely defined self-adjoint operators. These will in turn be used in the definition of the fractional Laplacian. A space of suitable test functions will also be defined for the fractional Laplacian.

Finally both discrete fractional Gaussian fields and fractional Gaussian fields are introduced along with their properties leading us to prove the convergence in distribution of discrete fractional Gaussian fields to fractional Gaussian fields on a space of suitable test functions.

We conclude the thesis with simulations of a discrete fractional Gaussian field on the Vicsek fractal.

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Chapter 1

Introduction

The notion of fractals was introduced in the 1970's by Benoît B. Mandelbrot, where he defined it as an object that has Hausdorff dimension strictly larger than its topological dimension. Many have since built upon this definition of fractals and greatly expanded the area of fractal analysis [5], [8], [10], [12]. The thesis will expand upon this by introducing and studying a family of Gaussian fields, their discrete analogues and the convergence in distribution of discrete fractional Gaussian fields on the Vicsek set fractal.

The thesis is an extension of the paper [23] by Fabrice Baudoin and Li Chen. It will follow the aforementioned paper, but will diverge slightly by studying the Vicsek fractal rather than the Sierpinski gasket.

The Vicsek fractal, which henceforth shall be referenced to by the Vicsek set, is defined as the non-empty compact set K such that

$$K = \bigcup_{i=1}^5 \mathcal{S}_i(K), \quad \text{where} \quad \mathcal{S}_i(x) = \frac{1}{3}(x - p_i) + p_i, \quad i = 1, 2, \dots, 5.$$

Here $p_i \in V_0$ are the edge points of a "plus" with a point in the middle, see Figure 1.1. The discrete Vicsek set is then defined recursively as

$$V_m = \bigcup_{i=1}^5 \mathcal{S}_i(V_{m-1}).$$

We can then create a discrete graph of the Vicsek set as shown in Figure 1.1.

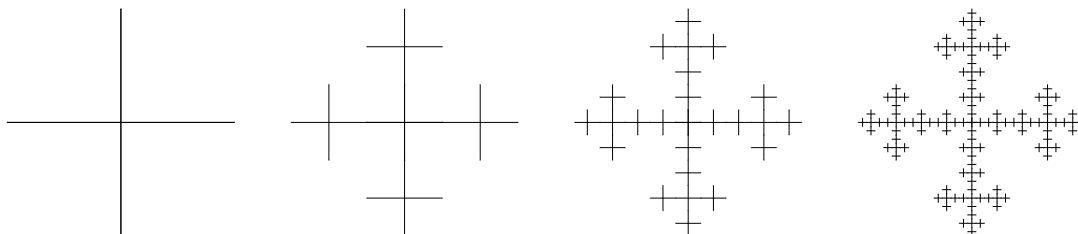


Figure 1.1: Vicsek graph, G_0, G_1, G_2 and G_3 .

As stated, the purpose of this thesis is to study the fractional Gaussian fields. So if we let $K \subset \mathbb{R}^2$ be the Vicsek set, then the family of fractional Gaussian fields we wish to study are defined on K , indexed by a parameter $s \geq 0$, satisfying

$$\mathbb{E}[X_s(f)X_s(g)] = \int_K (-\Delta)^{-s} f (-\Delta)^{-s} g \, d\mu.$$

Where f, g are in a space of suitable test functions on K , μ is a normalized Hausdorff measure and Δ is the Dirichlet Laplacian on K . Such fields are heuristically defined as the distribution $X_s = (-\Delta)^{-s} W$ where W is a white noise on $L^2(K, \mu)$.

Likewise the discrete fractional Gaussian fields are heuristically given by $X_s^m = (-\Delta_m)^{-s} W_i$ where Δ_m is the discrete Laplacian and W_i is a sequence of i.i.d. Gaussian random variables with mean 0 and variance 1. These are precisely the fields of which we will prove convergence in distribution.

As we will be studying the Vicsek set, it is natural that we start by constructing the set and proving some of its basic properties. Since we have described the Vicsek set as a non-empty compact self-similar set, we will first define and prove some basic properties of self-similar sets.

1.1 Iterated Function System

The iterated functions systems (IFS) gives us a way to work with self-similar sets. Constructing the IFS also allows us to prove that the Vicsek set actually exists. Thus we start by defining the iterated function systems.

Definition 1.1: A finite family of functions, $\{\mathcal{S}_1, \mathcal{S}_2, \dots, \mathcal{S}_m\}$, with $m \geq 2$ is called an *iterated function system* (IFS) if there exists $\{c_i\}_{1 \leq i \leq m} \in (0, 1)$ such that all $\mathcal{S}_i$ are *contractions*, i.e.

$$|\mathcal{S}_i(x) - \mathcal{S}_i(y)| \leq c_i |x - y|.$$

Furthermore we call a non-empty compact subset K of $\mathbb{R}^n$ an *invariant set* for the IFS if

$$K = \bigcup_{i=1}^m \mathcal{S}_i(K). \quad (1.1)$$

This then begs the question, if there for every IFS exists an invariant set. We shall see that this is indeed true, but to be able to prove that an invariant set exists, which is in fact unique, we will need to define the Hausdorff metric.

Definition 1.2: Define $A_\delta = \{x \in \mathcal{A} : \exists a \in A, |x - a| < \delta\}$. The *Hausdorff metric* is then given by

$$d_H(X, Y) = \inf \{\delta : X \subset Y_\delta \text{ and } Y \subset X_\delta\}.$$

The proof that d_H is a metric can be found in [11].

Theorem 1.3: Consider the IFS given by $\{\mathcal{S}_1, \mathcal{S}_2, \dots, \mathcal{S}_m\}$, then there exists a unique invariant set K for the IFS. Moreover, if we define a transformation S , on the class $\mathcal{A}$ of non-empty compact sets, as

$$S(E) = \bigcup_{i=1}^m \mathcal{S}_i(E), \quad \text{for } E \in \mathcal{A},$$

and denote by S^k the k 'th iterate of S , so $S^0 = E$ and $S^k(E) = S(S^{k-1}(E))$ for $k \geq 1$, then

$$K = \bigcap_{i=0}^{\infty} S^i(E)$$

for every set $E \in \mathcal{A}$, such that $E \supset \mathcal{S}_i(E)$ for all i .

Proof. Let E be an arbitrary set in $\mathcal{A}$ such that $\mathcal{S}_i(E) \subset E$ for all i .

Since $S^0 = E \supset S(E) = S^1(E)$, we find, by the induction assumption $S^k \subset S^{k-1}$, that $S^{k+1}(E) = S(S^k(E)) \subset S(S^{k-1}(E)) = S^k(E)$, in which case we get $S^k(E) \subset S^{k-1}(E)$. This in turn implies that $S^k(E)$ is a decreasing sequence of non-empty compact sets, in which case we have a non-empty compact intersection $K = \bigcap_{k \geq 0} S^k(E)$. Since $S^k(E)$ is a decreasing sequence, then $S(K) = K$ satisfying (1.1).

Now moving on to uniqueness, we will need the following inequality

$$\begin{aligned} d_H(S(A), S(B)) &= d_H\left(\bigcup_{i=1}^m \mathcal{S}_i(A), \bigcup_{i=1}^m \mathcal{S}_i(B)\right) \\ &\leq \max_{1 \leq i \leq m} d_H(\mathcal{S}_i(A), \mathcal{S}_i(B)) \leq \max_{1 \leq i \leq m} c_i \cdot d_H(A, B). \end{aligned} \quad (1.2)$$

Assuming A and B are both invariant sets of the same IFS, then $S(A) = A$ and $S(B) = B$, giving us that

$$d_H(A, B) = d_H(S(A), S(B)) \leq \max_{1 \leq i \leq m} c_i \cdot d_H(A, B).$$

Since $0 < \max_{1 \leq i \leq m} c_i < 1$, then $d_H(A, B) = 0$ with which $A = B$ making the invariant set unique.

Finally we will prove (1.2). The first inequality comes by choosing

$$\delta = \max_{1 \leq i \leq m} d_H(\mathcal{S}_i(A), \mathcal{S}_i(B))$$

since it follows that $\mathcal{S}_i(B) \subset (\mathcal{S}_i(A))_\delta$ for all $0 \leq i \leq m$. Thus it will definitely hold for $\bigcup_{i=1}^m \mathcal{S}_i(B) \subset (\bigcup_{i=1}^m \mathcal{S}_i(A))_\delta$ and vice versa.

The second inequality comes from letting $\delta = d_H(A, B)$. Then for every $a \in A$ we can choose some $b \in B$ such that $|a - b| \leq \delta$ so that $|\mathcal{S}_i(a) - \mathcal{S}_i(b)| \leq c_i \delta$ hence $\mathcal{S}_i(A) \subset (\mathcal{S}_i(B))_{c_i \delta}$. $\square$

Remark 1.4: A consequence of Theorem 1.3, is that we get a way to create a sequence $(i_1, i_2, \dots)$, which is not necessarily unique, where $1 \leq i_j \leq m$, such that for each $x \in F$ we can write

$$\{x\} = \{x_{i_1, i_2, \dots}\} = \bigcap_{k \geq 0} K_{i_1, \dots, i_k} \quad (1.3)$$

where $K_{i_1, \dots, i_k} = (\mathcal{S}_{i_1} \circ \dots \circ \mathcal{S}_{i_k})(K)$. Thus in a way we give each $x \in K$ an address. This is possible since S^k decreases as k increases.

If we further let $\mathcal{I}$ denote all sequences of the type $(i_1, i_2, \dots)$, then we can describe K as

$$K = \bigcup_{\mathcal{I}} \{x_{i_1, \dots, i_k}\}. \quad (1.4)$$

1.2 Construction of the Vicsek set

Let $V_0 = \{p_1, p_2, p_3, p_4, p_5\}$ be the set of vertices in $\mathbb{R}^2$ given by $p_1 = (0, 0)$, $p_2 = (\xi, 0)$, $p_3 = (0, -\xi)$, $p_4 = (-\xi, 0)$ and $p_5 = (0, \xi)$ for some $\xi > 0$. Then define

$$\mathcal{S}_i(z) = \frac{1}{3}(z - p_i) + p_i, \quad \text{for } i = 1, 2, \dots, 5$$

where $\{\mathcal{S}_1, \dots, \mathcal{S}_5\}$ is indeed an IFS, since for every $x, y \in \mathbb{R}^2$ we have

$$|\mathcal{S}_i(x) - \mathcal{S}_i(y)| = \left| \frac{1}{3}(x - p_i) + p_i - \left(\frac{1}{3}(y - p_i) + p_i \right) \right| = \frac{1}{3}|x - y|.$$

Thus by Theorem 1.3 we define the Vicsek set K as the unique non-empty compact subset in $\mathbb{R}^2$ such that

$$K = \bigcup_{i=1}^5 \mathcal{S}_i(K).$$

Since we are going to study the convergence of discretely defined operators we can define a discrete Vicsek set in the following way. We start by defining a sequence of sets $(V_m)_{m \geq 0}$ inductively:

$$V_{m+1} = \bigcup_{i=1}^5 \mathcal{S}_i(V_m).$$

This gives us a sequence of Vicsek set graphs $(G_m)_{m \geq 0}$ whose vertices are $(V_m)_{m \geq 0}$, as seen in Figure 1.1.

Notation: We denote $V^* = \bigcup_{m \geq 0} V_m$ and for any $p \in V_m$ we write the collection of neighbours of p in G_m as $V_{m,p}$. We may write $x \sim_m y$ when x and y are neighbours in V_m . We may also write $x \sim y$ when x and y are neighbours and there is no confusion as to what V_m x and y belongs to. Lastly, note that $N_m = \#V_m = 5^m \cdot 4 + 1$.

1.3 Hausdorff Dimension

The Hausdorff dimension can be thought of as a form of "fractal dimension" and will be important as it will be used to bound and define many of the mathematical objects that we will manipulate.

The Hausdorff dimension has the advantage of being defined for any set and that it is based on measures, which we know how to work with. The greatest disadvantage is that it is, in most cases, very difficult to calculate from the definition. If we are given an invariant set of an IFS, it will be quite easy to calculate. We begin this section by defining a cover for sets.

Definition 1.5: Let X be a metric space with the associated metric $d(\cdot, \cdot)$. We say the countable collection of sets, $\{U_i\}_{i \geq 1}$, is a δ -cover of K , if $K \subset \bigcup_{i \geq 1} U_i$ and $\text{diam } U_i \leq \delta$, where $\text{diam}(A) = \sup_{x, y \in A} d(x, y)$.

Definition 1.6: Let X be a metric space. If $K \subset X$ and $s \in [0, \infty)$ we define

$$H_\delta^s(K) = \inf \left\{ \sum_{i=1}^{\infty} \text{diam}(U_i)^s : \bigcup_{i=1}^{\infty} U_i \supseteq K, \text{diam}(U_i) < \delta \right\}.$$

The s -dimensional *Hausdorff measure* is then given by

$$H^s(K) = \lim_{\delta \rightarrow 0} H_\delta^s(K).$$

Remark 1.7: The Hausdorff measure defined above, is not a measure in the normal measure theory sense. Rather it is an outer measure, although it is possible to prove that the Hausdorff measure restricted to Borel-sets, is indeed a measure.

Moving swiftly along, we now want to investigate the behaviour of H^s as s changes. Let $\{U_i\}_{i \geq 1}$ be a δ -cover of K , then we have for $0 \leq s < t$ that

$$\sum_{i \geq 1} \text{diam}(U_i)^t = \sum_{i \geq 1} \text{diam}(U_i)^{t-s} \text{diam}(U_i)^s \leq \delta^{t-s} \sum_{i \geq 1} \text{diam}(U_i)^s.$$

Taking infimum on both sides we get that $H_\delta^t(K) \leq \delta^{t-s} H_\delta^s(K)$. If we let $\delta \rightarrow 0$ we see that if $H^s(K) < \infty$ then $H^t(K) = 0$ since $\delta^{t-s} \rightarrow 0$. Likewise if $H^t > 0$ then $H^s = \infty$. This tells us that if there is a point, say κ , such that H^κ is both positive and finite, then H^s will be 0 for all $s > \kappa$ and ∞ for all $s < \kappa$. Such a value is called the Hausdorff Dimension of K .

Definition 1.8: The *Hausdorff dimension*, denoted $d_h(K)$ is given by

$$d_h(K) = \inf \{s \geq 0 : H^s(K) = 0\} = \sup \{s \geq 0 : H^s(K) = \infty\}.$$

Notation: When there is no confusion with regards to what set we are working with, we denote the Hausdorff dimension by d_h .

As stated before, it is quite difficult to determine the Hausdorff dimension from the definition. Fortunately, we can with a few assumptions find the Hausdorff dimension quite easily. But before we can prove this theorem, we first define the open set condition and state a few important lemmas.

Definition 1.9: We say that an IFS, $\{\mathcal{S}_i\}_{1 \leq i \leq m}$, satisfies the *open set condition* if there exists a non-empty bounded open set V so

$$V \supset \bigcup_{i=1}^m \mathcal{S}_i(V)$$

such that each $\mathcal{S}_i(V)$ is disjoint.

The following lemma describes a lower bound for the Hausdorff dimension.

Lemma 1.10: Let μ be a measure such that $\mu(F) \in (0, \infty)$ and suppose there are constants $c, \varepsilon > 0$ such that for some $s \geq 0$

$$\mu(U) \leq c \operatorname{diam}(U)^s$$

for all sets U with $\operatorname{diam}(U) \leq \varepsilon$. Then $H^s(F) \geq \frac{\mu(F)}{c}$ and $s \leq d_h$.

Proof. If $\{U_i\}_{i \in I}$ is any cover of F then

$$0 < \mu(F) \leq \mu\left(\bigcup_{i \in I} U_i\right) \leq \sum_{i \in I} \mu(U_i) \leq c \sum_{i \in I} \operatorname{diam}(U_i)^s$$

by definition of the measure μ and its properties. Taking infimum we get $H_\delta^s \geq \frac{\mu(F)}{c}$ and therefore $H^s \geq \frac{\mu(F)}{c} > 0$ so by Definition 1.8 $s \leq d_h$. $\square$

Lemma 1.11: Let $\{V_i\}$ be a collection of disjoint open subsets of $\mathbb{R}^n$ such that each V_i contains a ball of radius $a_1 r$ and is contained in a ball of radius $a_2 r$. Then any ball B of radius R intersects at most $(1 + 2a_2)^n a_1^n$ of the closures $\bar{V}_i$

Proof. By assumption we know that $\operatorname{diam}(V_i) \leq 2a_2 r$ for all V_i . So if some $\bar{V}_i$ intersects the ball with radius r , then $\bar{V}_i$ must be contained in the ball with radius $(2a_2 + 1)r$ concentric with the ball with radius r . Now let q denote the number of closures $\bar{V}_i$ such that $\bar{V}_i \cap B(x, r) \neq \emptyset$. Then there exists q disjoint balls with radius $a_1 r$ that are contained in the ball with radius $(2a_2 + 1)r$, since by assumption, each $\bar{V}_i$ contains a ball of radius $a_1 r$. It follows that $q(a_1 r)^n \leq (2a_2 + 1)^n r^n$ which gives us the bound for q . $\square$

Combing the above results, we are able to state the following theorem:

Theorem 1.12: Suppose Definition 1.9 holds for the similarities $\{\mathcal{S}_i\}_{1 \leq i \leq m}$ on $\mathbb{R}^n$ with ratios $0 < c_i < 1$. If, for K the following holds

$$K = \bigcup_{i=1}^m \mathcal{S}_i(K) \tag{1.5}$$

then $d_h = s$ where s is given by

$$\sum_{i=1}^m c_i^s = 1. \tag{1.6}$$

Moreover, for this value of s , $0 < H^s(K) < \infty$.

Proof. Let $\mathcal{I}_k$ denote the set of all k -term sequences $(i_1, \dots, i_k)$ with $1 \leq i_j \leq m$. For any set A and any sequence $(i_1, \dots, i_k) \in \mathcal{I}_k$ we denote $A_{i_1, \dots, i_k} = (\mathcal{S}_{i_1} \circ \dots \circ \mathcal{S}_{i_k})(A)$.

We will start by finding an upper bound for the Hausdorff measure. We do this by finding an appropriate δ -cover for K . It follows that by repeated use of (1.5) that

$$K = \bigcup_{i=1}^m \mathcal{S}_i(K) = \bigcup_{i=1}^m \mathcal{S}_i \left(\bigcup_{j=1}^m \mathcal{S}_j(K) \right) = \bigcup_{i,j=1}^m (\mathcal{S}_i \circ \mathcal{S}_j)(K) = \dots = \bigcup_{\mathcal{I}_k} \mathcal{S}_{i_1, \dots, i_k}(K) = \bigcup_{\mathcal{I}_k} K_{i_1, \dots, i_k}.$$

Now noting that $\mathcal{S}_{i_1} \circ \dots \circ \mathcal{S}_{i_k}$ has the ratios $c_{i_1} \dots c_{i_k}$, we see

$$\sum_{\mathcal{I}_k} \text{diam}(K_{i_1, \dots, i_k})^s = \sum_{\mathcal{I}_k} (c_{i_1} \dots c_{i_k})^s \text{diam}(K)^s = \left(\sum_{i_1} c_{i_1}^s \right) \dots \left(\sum_{i_k} c_{i_k}^s \right) \text{diam}(K)^s = \text{diam}(K)^s$$

where the last equality comes from (1.6). Now for any $\delta > 0$ we can choose k so

$$\text{diam}(K_{i_1, \dots, i_k}) \leq \left(\max_{1 \leq i \leq m} c_i \right)^k \text{diam}(K) \leq \delta$$

since $\max_{1 \leq i \leq m} c_i \in (0, 1)$. Thus $\{K_{i_1, \dots, i_k}\}_{\mathcal{I}_k}$ is a δ -cover for K whereby $H_\delta^s(K) \leq \text{diam}(K)^s$ in which case $H^s(K) \leq \text{diam}(K)^s$ giving an upper bound.

Next we need a lower bound for the Hausdorff dimension, which is substantially more difficult. Let $\mathcal{I} = \{(i_1, i_2, \dots) : 1 \leq i_j \leq m\}$ be the set of all infinite sequences and let

$$\mathcal{I}_{i_1, \dots, i_k} = \{(i_1, \dots, i_k, q_{k+1}, \dots) : 1 \leq q_j \leq m\},$$

so the first initial k "terms" are chosen and the rest are the remaining permutations.

We now create an outer measure μ on a subset of $\mathcal{I}$ defined by $\mu(\mathcal{I}_{i_1, \dots, i_k}) = (c_{i_1} \dots c_{i_k})^s$. From this outer measure μ we construct an outer measure on K named $\tilde{\mu}$ defined in the following way. For some subset $A \subseteq K$

$$\tilde{\mu}(A) = \mu \left(\{(i_1, i_2, \dots) \in \mathcal{I} : x_{i_1, i_2, \dots} \in A \cap K\} \right).$$

The proof that μ and $\tilde{\mu}$ are indeed outer measures can be found in [25, Section 7].

We want to show that $\tilde{\mu}$ fulfils the properties of Lemma 1.10.

Let V be an open set that satisfies Definition 1.9. Since $\bar{V} \supset \mathcal{S}(\bar{V}) = \bigcup_{i=1}^m \mathcal{S}_i(\bar{V})$, then by Theorem 1.3 the decreasing sequence $S^k(\bar{V})$ converges to K . In particular we have $\bar{V} \supset S(\bar{V}) \supseteq \bigcup_{i=1}^m \mathcal{S}_i(\bar{V}) = K$ therefore

$$\bar{V}_{i_1, \dots, i_k} = (\mathcal{S}_{i_1} \circ \dots \circ \mathcal{S}_{i_k})(\bar{V}) \supseteq (\mathcal{S}_{i_1} \circ \dots \circ \mathcal{S}_{i_k})(K) = K_{i_1, \dots, i_k},$$

for each finite sequence $(i_1, \dots, i_k)$.

Now let B be any ball $r < 1$. We estimate $\tilde{\mu}(B)$ by considering sets $V_{i_1, \dots, i_k}$, that have diameters comparable with that of B , with closures intersecting $K \cap B$. Further we define $\mathcal{Q}$ as the set of sequences $(i_1, \dots, i_k) \in \mathcal{I}$ such that i_k is the smallest k for which the following holds

$$\left(\min_{1 \leq i \leq m} c_i \right) r \leq c_{i_1} \dots c_{i_k} \leq r.$$

Note that no sequence in $\mathcal{Q}$ extends to another sequence in $\mathcal{Q}$. Since, if two sequences are the same up until a certain point, and one is longer than the other, then the shorter sequence would be the one in $\mathcal{Q}$. Now by Definition 1.9 $V_1, \dots, V_m$ are disjoint and therefore so are $V_{i_1, \dots, i_k, 1}, \dots, V_{i_1, \dots, i_k, m}$ for each $(i_1, \dots, i_k)$. In particular we have that $\{V_{i_1, \dots, i_k} : (i_1, \dots, i_k) \in \mathcal{Q}\}$ is a collection of disjoint sets. We further note that by (1.4) we have

$$K = \bigcup_{\mathcal{I}} \{x_{i_1, i_2, \dots}\} \subseteq \bigcup_{\mathcal{Q}} K_{i_1, \dots, i_k} \subseteq \bigcup_{\mathcal{Q}} \bar{V}_{i_1, \dots, i_k}. \quad (1.7)$$

Next we choose a_1 and a_2 so that V contains a ball of radius a_1 and is contained in a ball of radius a_2 . Then for all $(i_1, \dots, i_k) \in \mathcal{Q}$ the set $V_{i_1, \dots, i_k}$ contains a ball of radius $c_{i_1} \dots c_{i_k} a_1$ and therefore one of radius $(\min_{1 \leq i \leq m} c_i) a_1 r$. Furthermore it is contained in a ball of radius $c_{i_1} \dots c_{i_k} a_2$ and therefore one of radius $a_2 r$.

Now let $\mathcal{Q}_B$ be the set of all sequences $(i_1, \dots, i_k) \in \mathcal{Q}$ such that $\bar{V}_{i_1, \dots, i_k}$ intersects B . By Lemma 1.11 there are at most

$$q = (1 + 2b)^n \left(\min_{1 \leq i \leq m} c_i \right)^{-n} a^{-n}$$

sequences in $\mathcal{Q}_B$. Thus we find that if

$$x_{i_1, \dots, i_k} \in K \cap B \subseteq \bigcup_{\mathcal{Q}_B} \bar{V}_{i_1, \dots, i_k}, \quad \text{by (1.7),}$$

then there must be some k such that $(i_1, \dots, i_k) \in \mathcal{Q}_B$ hence $(i_1, i_2, \dots) \in \bigcup_{\mathcal{Q}_B} \mathcal{I}_{i_1, \dots, i_k}$. We can therefore conclude

$$\tilde{\mu}(B) = \mu \left(\{(i_1, i_2, \dots) \in \mathcal{I} : x_{i_1, i_2, \dots} \in K \cap B\} \right) \leq \mu \left(\bigcup_{\mathcal{Q}_B} \mathcal{I}_{i_1, \dots, i_k} \right),$$

whereby it follows that

$$\tilde{\mu}(B) \leq \sum_{\mathcal{Q}_B} \mu(\mathcal{I}_{i_1, \dots, i_k}) = \sum_{\mathcal{Q}_B} (c_{i_1} \dots c_{i_k})^s \leq \sum_{\mathcal{Q}_B} r^s \leq q r^s.$$

Since any set U is contained in a ball of radius $\text{diam}(U)$, we have that $\tilde{\mu} \leq q \text{diam}(U)^s$. By Lemma 1.10 we have that $H^s(K) \geq \frac{\tilde{\mu}(K)}{q} = q^{-1} > 0$, thus $0 < q^{-1} \leq H^s(K) \leq \text{diam}(K)^s < \infty$ such that by Definition 1.8 we have that $s = d_h$. $\square$

1.3.1 Spectral and Walk dimension

Later in this thesis we will bound certain random fields and functions by a constant we call the spectral dimension which depends on two factors, namely the Hausdorff dimension, which we have just discussed, and the walk dimension. Properly explaining the walk dimension is outside the scope of this thesis, but we will give some intuition as to what it is.

Let B_t be a Brownian motion on K , and $B(x, r)$ be the open ball centered at $x \in K$ with radius r . Then the walk dimension is the constant $d_w > 0$ such that

$$c_1 r^{d_w} \leq \mathbb{E} [\tau(x, r)] \leq c_2 r^{d_w},$$

where $\tau(x, r) = \inf \{t \geq 0 : B_t \notin B(x, r)\}$. The intuition is that it is the average escape rate, meaning how fast the Brownian motion escapes the ball. This explanation for the walk dimension is a rather simplified and condensed one. For further reference see [10, Chapter 3]. It does however, allow us to define the spectral dimension as follows:

Definition 1.13: Let d_h be the Hausdorff dimension and d_w the walk dimension then the *spectral dimension* is

$$d_s = \frac{2d_h}{d_w}.$$

1.4 Hausdorff, Walk and Spectral Dimension of the Vicsek set

Since the main purpose of the thesis is to study the Vicsek set, we will in this section, state the different dimensions needed.

Recalling that the Vicsek set is the unique set K for the IFS $\{\mathcal{S}_1, \dots, \mathcal{S}_5\}$ with $\mathcal{S}_i(z) = \frac{1}{3}(z - p_i) + p_i$ where $p_i \in V_0$, it holds that

$$K = \bigcup_{i=1}^5 \mathcal{S}_i(K).$$

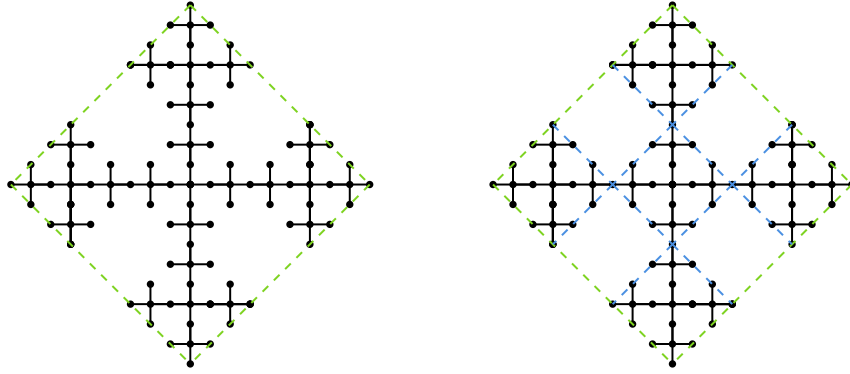
We want to use Theorem 1.12 to find the Hausdorff dimension of the Vicsek set, so we need to check that $\mathcal{S}_{i1 \leq i \leq 5}$ satisfies Definition 1.9.

If we choose the open set V to be the open square where each corner is a point in V_0 , as seen in Figure 1.2, then by continuity of $\mathcal{S}_i$ we have that $\mathcal{S}_i(V)$ is open and disjoint. The disjointness of $\mathcal{S}_i$ is that each $\mathcal{S}_i$ "retracts" the whole set towards the point p_i and we get that the sets are disjoint. One might worry about the points where the open sets meet, but since the points in V_0 , excluding the middle point, are not included in the initial open box, we have that they do not intersect.

We can therefore conclude from Theorem 1.12 that

$$\sum_{i=1}^5 \left(\frac{1}{3}\right)^{d_h} = 1 \Rightarrow \left(\frac{1}{3}\right)^{d_h} = \frac{1}{5} \Rightarrow d_h = \frac{\log 5}{\log 3}.$$

As mentioned earlier we will not be calculating the walk dimension, but we will state it. The walk dimension for the Vicsek set is $d_w = d_h + 1$ and we can state the following lemma.

Figure 1.2: The opens sets V (green) and $\mathcal{S}_i(V)$ (blue).

Lemma 1.14: The following constants hold for the Vicsek set K

$$d_h = \frac{\log 5}{\log 3}, \quad d_w = d_h + 1, \quad \text{and} \quad d_s = \frac{d_h}{2d_w} = \frac{\log 5}{2 \log 15}.$$

Having calculated the Hausdorff dimension of the Vicsek set we will immediately use it to construct the following measure.

Definition 1.15: The *normalized Hausdorff measure* on K is given by the Borel measure μ on K , such for any $(i_1, \dots, i_n)$ where $1 \leq i_j \leq 5$, we have

$$\mu(\mathcal{S}_{i_1} \circ \dots \circ \mathcal{S}_{i_n}(K)) = 5^{-n}.$$

The measure μ has the following property:

Proposition 1.16: The measure μ is d_h -Ahlfors regular, i.e. there exists $c_1, c_2 > 0$ such that for every $x \in K$ and $r \in [0, \text{diam}(K)]$ we have

$$c_1 r^{d_h} \leq \mu(B(x, r)) \leq c_2 r^{d_h}.$$

Proof. If $r = 0$ the proof is trivial. So let $r \in (0, 1)$, and choose n so $R \left(\frac{2}{3}\right)^{n+1} \leq r \leq R \left(\frac{2}{3}\right)^n$, where $R = \text{diam}(K)$. Note that for a sequence, $(i_1, \dots, i_m)$, we have

$$\text{diam}(\mathcal{S}_{i_1} \circ \dots \circ \mathcal{S}_{i_m}(K)) = \left(\frac{2}{3}\right)^m \cdot R.$$

We can therefore construct a sequence $i_1, \dots, i_n$ and $i_1, \dots, i_{n+1}$ such that

$$B(x, r) \supset (\mathcal{S}_{i_1} \circ \dots \circ \mathcal{S}_{i_{n+1}})(K), \quad \text{and} \quad (\mathcal{S}_{i_1} \circ \dots \circ \mathcal{S}_{i_n})(K) \cap B(x, r) \neq \emptyset.$$

Since $B(x, r)$ intersects with at most 5 of these sequences, we find an upper bound to be

$$\mu(B(x, r)) \leq 5 \cdot \mu((\mathcal{S}_{i_1} \circ \dots \circ \mathcal{S}_{i_n})(K)) = 5 \cdot 5^{-n} = 5^2 \cdot 5^{-(n+1)} \leq 5 \cdot (R \cdot 3^{-(n+1)})^{d_h} \leq 5 \cdot 2^{-n} \cdot r^{d_h}$$

noting that $2^{-n} r \leq r$.

This upper bound also works for $R < 1$ since we just omit the constant from the calculations and have $3^{-(n+1)} \leq R^{-1}2^{-(n+1)}r$ and the upper bound becomes $5 \cdot \left(R^{-1}2^{-(n+1)}r\right)^{d_h}$.

Likewise, for the lower bound we have that there exists a sequence $i_1, \dots, i_{n+1}$ such that $B(x, r) \supset S_{i_1} \circ \dots \circ S_{i_{n+1}}$ in which case

$$\mu(B(x, r)) \geq \mu\left(\left(S_{i_1} \circ \dots \circ S_{i_{n+1}}\right)(K)\right) = 5^{-(n+1)} \geq R \cdot 5^{-(n+1)} \geq 5 \cdot \left(R^{-1}2^{-n}r\right)^{d_h}.$$

Again this lower bound also holds for $R > 1$ since we again omit R in the calculations and have $R^{-1}2^{-n}r \leq 3^{-n}$ hence the lower bound becomes $5 \cdot \left(R^{-1}2^{-n}r\right)^{d_h}$.

Now assume $\text{diam}(K) \geq r \geq 1$. The upper bound is simply $\mu(B(x, r)) \leq \mu(K) = 1 \leq r$. For the lower bound, there exists $m \in \mathbb{N}$ so $3^{m-1} \leq r \leq 3^m$, such that

$$\mu(B(x, r)) \geq \mu(\mathcal{S}_{i_1} \circ \dots \circ \mathcal{S}_{i_{m-1}}) = 5^{-(m-1)} = 5 \cdot (3^{-m})^{d_h} \geq (r3^{-2m})^{d_h}$$

as desired. $\square$

As we will want to work with the graph approximations of the Vicsek set, we will define a sequence of discrete measures.

Definition 1.17: Let $(\mu_m)_{m \geq 0}$ be a sequence of discrete measures on V_m given by

$$\mu_m = \frac{1}{a_m} \sum_{p \in V_m} \delta_p, \quad (1.8)$$

where $a_m = 5^m \cdot 4 + 1$.

We have that $\mu_m \rightarrow \mu$ as $m \rightarrow \infty$. This follows from the fact that $\mathcal{S}_i(K)$ "shrinks" the fractal down to $1/5$ its size around the point p_i , in which case for any sequence $(i_1, \dots, i_n)$ we have that

$$\begin{aligned} \lim_{m \rightarrow \infty} \mu_m(\mathcal{S}_{i_1} \circ \dots \circ \mathcal{S}_{i_n}(K)) &= \lim_{m \rightarrow \infty} \frac{1}{a_m} \sum_{p \in V_m} \delta_p \left((\mathcal{S}_{i_1} \circ \dots \circ \mathcal{S}_{i_n})(K) \right) \\ &= \lim_{m \rightarrow \infty} \frac{1}{a_m} \cdot 5^{-n} \cdot (5^m \cdot 4 + 1) = 5^{-n} = \mu \left((\mathcal{S}_{i_1} \circ \dots \circ \mathcal{S}_{i_n})(K) \right) \end{aligned}$$

recalling that $\#V_m = 5^m \cdot 4 + 1$. This leads us to the following corollary:

Corollary 1.18: The sequence of measures $(\mu_m)_{m \geq 0}$ defined in (1.8) converges weakly to the normalized Hausdorff measure μ on the Vicsek set K i.e.

$$\lim_{m \rightarrow \infty} \int_K f \, d\mu_m = \int_K f \, d\mu.$$

Chapter 2

Laplacians

In Chapter 3 the fractional Gaussian fields will be defined using the eigenvalues and eigenfunctions of the Dirichlet Laplacian. We will find that the Dirichlet Laplacian is an extension of the Kigami Laplacian. But to construct the Kigami Laplacian we need the discrete Laplacian as we will prove that the Kigami Laplacian is the pointwise limit of the discrete Laplacian. We will therefore start by introducing the discrete Laplacian, then the Kigami Laplacian, culminating with the Dirichlet Laplacian. Therefore consider the discrete Laplacian on the Vicsek set.

Definition 2.1: Let $\ell(V_m)$ be the set of functions $f: V_m \rightarrow \mathbb{R}$. Then for any $f \in \ell(V_m)$ the *discrete Laplacian* on V_m is defined as

$$\Delta_m f(p) = 3^m \frac{1}{N_p} \sum_{q \in V_{m,p}} (f(q) - f(p)), \quad p \in V_m \setminus V_0$$

where

$$N_p = \begin{cases} 5^{-m}, & \text{if } \#V_{m,p} = 4 \\ 2 \cdot 5^{-(m+1)}, & \text{if } \#V_{m,p} = 2 \\ 5^{-(m+1)}, & \text{if } \#V_{m,p} = 1. \end{cases}$$

One is able to describe the discrete Laplacian of a function f as a matrix called the Laplacian matrix.

Definition 2.2: Let $p_1, \dots, p_{N_m}$ be the vertices in V_m then the *Laplacian matrix* L_m is given by

$$(L_m)_{i,j} = \begin{cases} \#V_{m,p_i} \cdot 3^m \cdot \frac{1}{N_{p_i}}, & \text{if } i = j \\ -1 \cdot 3^m \cdot \frac{1}{N_{p_i}}, & \text{if } i \neq j \text{ and } p_j \in V_{m,p_i} \\ 0, & \text{otherwise} \end{cases}$$

which satisfies:

$$L_m F = \begin{pmatrix} \Delta_m f(x_1) \\ \vdots \\ \Delta_m f(x_{N_m}) \end{pmatrix}, \quad \text{where } F = \begin{pmatrix} f(x_1) \\ \vdots \\ f(x_{N_m}) \end{pmatrix}.$$

2.1 Graph Energies

In this section we will be following the arguments given by Strichartz in [17, Chapter 1]. Many of the proofs are going to rely on the graph energy. Thus, we will first define and prove some properties of the graph energies.

Definition 2.3: Given a finite connected graph G_m and its vertices V_m then for functions $u, v: V_m \rightarrow \mathbb{R}$ we define the *graph energy* to be

$$E_m(u, v) = \sum_{x \sim y} (u(x) - u(y)) (v(x) - v(y)).$$

Notation: We will write $E_m(u) = E_m(u, u)$.

Since we want to work on the entire Vicsek set K , we will want to create an extension of a function $u: V_m \rightarrow \mathbb{R}$, such that it is defined on V_{m+n} . This is done in the following way:

Definition 2.4: Let $u: V_m \rightarrow \mathbb{R}$. Then an *extension of u* is a function $u': V_{m+n} \rightarrow \mathbb{R}$ where $u' \upharpoonright_{V_m} = u$.

One quickly notices that there must be a function that minimizes the graph energy of a given graph. We denote such a function by $\tilde{u}$ and call it a *harmonic extension*.

Definition 2.5: Let $\tilde{u}: V_{m+n} \rightarrow \mathbb{R}$ with $\tilde{u} \upharpoonright_{V_m} = u$ being a *harmonic extension* if $E_{m+n}(\tilde{u}) \leq E_{m+n}(u')$ for all $u': V_{m+n} \rightarrow \mathbb{R}$ with $u' \upharpoonright_{V_m} = u$.

In some cases we have that $E_{m+n}(\tilde{u}) = r^n E_m(u)$ for some $r \in (0, 1)$. But instead of normalizing the values on E_m we will normalize E_{m+n} , thus getting

$$r^{-n} E_{m+n}(\tilde{u}) = E_m(u), \quad \text{and} \quad r^{-n} E_{m+n}(u') \geq E_m(u)$$

for every extension u' of u . The constant r is called the *renormalization constant*. From the equality above we can define the following.

Definition 2.6: The *renormalized graph energies* are defined as

$$\mathcal{E}_m(u) = r^{-m} E_m(u).$$

Note that the sequence $\{\mathcal{E}_m(u)\}_{m \geq 0}$ is a non-decreasing sequence.

Based on this definition we see that for a harmonic extension $\tilde{u}$ of u we have that

$$\mathcal{E}_{m+1}(\tilde{u}) = r^{-(m+1)} E_{m+1}(\tilde{u}) = r^{-m} E_m(u) = \mathcal{E}_m(u).$$

Extending this to the bilinear form we have, for harmonic extension $\tilde{u}, \tilde{v}$ of u, v that

$$\mathcal{E}_{m+1}(\tilde{u}, \tilde{v}) = \mathcal{E}_m(u, v).$$

It is also true for the bilinear form, that if $\tilde{u}$ is the harmonic extension of u and v' is any extension of v then

$$\mathcal{E}_{m+1}(\tilde{u}, v') = \mathcal{E}_m(u, v). \tag{2.1}$$

For further details see [17].

Now that we have Definition 2.6, it is possible to define a harmonic function.

Definition 2.7: A function u is a *harmonic function* if u minimizes $\mathcal{E}_m(u)$ for all m , given boundary values on V_0 .

The harmonic functions will be used extensively throughout this section. We start by finding the renormalization constant for the Vicsek set.

2.1.1 The Renormalization Constant on the Vicsek set

In the previous section we talked about a harmonic function, but without ever constructing one. We will therefore in this section show how to construct a harmonic function for the Vicsek set for any initial boundary values $u(V_0)$.

Given $V_0 = \{q_1, q_2, q_3, q_4, q_5\}$ and $u: V_0 \rightarrow \mathbb{R}$ we have that

$$E_0(u) = (u(q_1) - u(q_2))^2 + (u(q_1) - u(q_3))^2 + (u(q_1) - u(q_4))^2 + (u(q_1) - u(q_5))^2.$$

We are going to exploit the symmetry of the Vicsek set. Since all four "extremities" of the Vicsek set are identical, as seen in Figure 2.1, we will only look at one of the extremities, since the other extremities will be the same computations. So if we want to minimize $E_1(u')$, for $u': V_1 \rightarrow \mathbb{R}$, we see for the extremity between q_1 and q_2 that

$$\begin{aligned} E_1(u') = & \left(u'(q_2) - u'(p_1)\right)^2 + \left(u'(p_1) - u'(p_3)\right)^2 + \left(u'(p_1) - u'(p_4)\right)^2 \\ & + \left(u'(p_1) - u'(p_5)\right)^2 + \left(u'(p_4) - u'(q_1)\right)^2. \end{aligned} \quad (2.2)$$

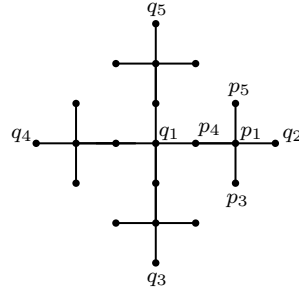


Figure 2.1: Points on the Vicsek set V_1 .

Next we are going to minimize the graph energies with regards to $u'(p_i)$ for $i = 0, 1, 2, 4$. Taking the derivative of the right side of (2.2) and solving for 0 we get the following

$$\begin{aligned} 4u'(p_1) &= u'(q_2) + u'(p_3) + u'(p_4) + u'(p_5) \\ 2u'(p_4) &= u'(q_1) + u'(p_1) \\ u'(p_3) &= u'(p_1) \\ u'(p_5) &= u'(p_1). \end{aligned}$$

Solving these equations we get

$$\begin{aligned} u'(p_1) &= \frac{2}{3}u'(q_2) + \frac{1}{3}u'(q_1) \\ u'(p_3) &= \frac{2}{3}u'(q_2) + \frac{1}{3}u'(q_1) \\ u'(p_5) &= \frac{2}{3}u'(q_2) + \frac{1}{3}u'(q_1) \\ u'(p_4) &= \frac{2}{3}u'(q_1) + \frac{1}{3}u'(q_2). \end{aligned}$$

Ergo what we have found is, that to minimize the value in the points of the Vicsek set, the value in the next point is going to be $2/3$ the value from the closest point and $1/3$ the value from the second closest point.

If we calculate the graph energy we find that

$$E_1(u') = \frac{1}{3} \sum_{i=1}^4 (u'(q_i) - u'(q_1))^2 = \frac{1}{3} E_0(u').$$

Since the Vicsek set is a self-similar set, we have that going from V_1 to V_2 is the same as going from V_m to V_{m+1} . Thus we can simply extend the above calculations repeating it m times, in which case

$$E_{m+1}(u') = \frac{1}{3} E_m(u').$$

We then get our renormalization constant for the Vicsek set to be $r = \frac{1}{3}$. Altogether we find from Definition 2.6 that

$$\mathcal{E}_m(u) = 3^m E_m(u).$$

2.2 The Kigami Laplacian

With the renormalization constant found, we will return to finding the Kigami Laplacian. Recall from Definition 2.6 that we have defined a sequence $\mathcal{E}_m$ for functions on V^* . It will therefore make sense to define the following limit.

Definition 2.8: Let $u: V^* \rightarrow \mathbb{R}$ then define the limit

$$\mathcal{E}(u, u) = \lim_{m \rightarrow \infty} \mathcal{E}_m(u, u),$$

where we set

$$\text{dom } \mathcal{E} = \{u \in \ell(V_*) : \mathcal{E}(u) < \infty\}, \quad \text{and} \quad \text{dom}_0 \mathcal{E} = \{u \in \text{dom } \mathcal{E} : u = 0, \text{ on } V_0\}.$$

This definition only involves functions defined on V^* , but we find that functions in $\text{dom } \mathcal{E}$ are uniformly continuous and therefore have a unique extension to K . We even have that $\text{dom } \mathcal{E}$ is dense in $C(K)$.

Proposition 2.9: Let $\text{dom } \mathcal{E}$ be defined as in Definition 2.8. Then the following holds

- (i) Let $u \in \text{dom } \mathcal{E}$, then u is uniformly continuous on V^* .
- (ii) $\text{dom } \mathcal{E}$ is dense in $C(K)$.

Proof. We start with (i). Let $u \in \text{dom } \mathcal{E}$ then $\mathcal{E}_m(u) \leq \mathcal{E}(u)$ since $\mathcal{E}_m$ is non-decreasing in m . So for any $x, y \in V_m$ if $y \in V_{m,x}$ we have that

$$r^{-m}(u(x) - u(y))^2 \leq \mathcal{E}_m(u) \leq \mathcal{E}(u) \tag{2.3}$$

since $r^{-m}(u(x) - u(y))^2$ is a term in $\mathcal{E}_m(u)$. Thus

$$|u(x) - u(y)| \leq r^{m/2} \mathcal{E}(u)^{1/2}.$$

Now consider a chain of points $x_m, x_{m+1}, \dots, x_{m+k}$ where $x_{m+j} \in V_{m+j}$ and $x_{m+j} \in V_{m+j+1, x_{m+j+1}}$. Then we have

$$|u(x_m) - u(x_{m+k})| \leq r^{m/2} \left(1 + r^{1/2} + \dots + r^{k/2}\right) \mathcal{E}(u)^{1/2} \leq \frac{r^m}{1 - r^{1/2}} \mathcal{E}^{1/2}$$

by using (2.3) and the triangle inequality multiple times. From the geometry of K we see that $x, y \in V^*$ belong to the same or adjacent $(\mathcal{S}_{i_1} \circ \dots \circ \mathcal{S}_{i_m})(K)$. We can then connect x to y by one of such chains, so

$$|u(x) - u(y)| \leq \frac{2r^{m/2}}{1 - r^{1/2}} \mathcal{E}(u)^{1/2}$$

giving a bound for all $x, y \in V^*$.

For proof of (ii) we refer to [17]. □

We are now ready to define the Kigami Laplacian. We will denote this with Δ_μ as it depends on the choice of measure μ . In our case we choose μ to be given as in Definition 1.15.

Definition 2.10: Let $u \in \text{dom } \mathcal{E}$ and let $f \in C(K)$, then $u \in \text{dom } \Delta_\mu$ and $\Delta_\mu u = f$ if

$$\mathcal{E}(u, v) = \int_K f v \, d\mu, \quad \forall v \in \text{dom}_0 \mathcal{E}.$$

We call Δ_μ the *Kigami Laplacian*.

As mentioned before, we want Δ_μ to be defined as the limit of Δ_m . The constants 3^m and 5^m are not random, but are chosen such that the Kigami Laplacian makes sense. But to prove the pointwise limit and actually calculate the constants, we will need a few lemmas and definitions.

We will introduce a new notation for the discrete graph Laplacian without its constants. The reason for this will become apparent, since we in future proofs will see that the constants cancels each other out. We start by defining this graph Laplacian followed by a harmonic function which will be used extensively in the following proofs.

Definition 2.11: Let Δ_m be defined as in Definition 2.1. We then define $\tilde{\Delta}_m$ as

$$\tilde{\Delta}_m f(p) = 3^{-m} N_p \Delta_m f(p) = \sum_{q \in V_{m,p}} f(q) - f(p).$$

Definition 2.12: Given $m \geq 0$ and a point $x \in V_m \setminus V_0$ then define the *harmonic function* $\psi_x^{(m)}: V^* \rightarrow [0, 1]$ that for a $y \in V_m$ is $\psi_x^{(m)}(y) = \delta_{xy}$.

Note that $\psi_x^{(m)} \in \text{dom } \mathcal{E}$. Furthermore, it might seem odd that we only defined $\psi_x^{(m)}$ to have values on V_m . However the function is defined on the whole of V^* , where the values in $K \setminus V_m$ come from the fact that it is a harmonic extension. We even have that function $\psi_x^{(m)}$ can be used to describe $\tilde{\Delta}_m$.

Proposition 2.13: Let $u \in \text{dom } \mathcal{E}$ and $\Delta_\mu u = f$ then

$$r^{-m} \tilde{\Delta}_m u(x) = \int_K f \psi_x^{(m)} \, d\mu.$$

Proof. We see that

$$\begin{aligned}\mathcal{E}_m(u, \psi_x^{(m)}) &= r^{-m} \sum_{p \sim_m q} (u(q) - u(p)) (\psi_x^{(m)}(p) - \psi_x^{(m)}(q)) \\ &= r^{-m} \sum_{q \in V_{m,x}} (u(x) - u(q)) = r^{-m} \tilde{\Delta}_m u(x),\end{aligned}$$

where the second equality stems from the fact that only edges containing x are going to see change in $\psi_x^{(m)}$. By (2.1) we have that

$$r^{-m} \tilde{\Delta}_m u(x) = \mathcal{E}_m(u, \psi_x^{(m)}) = \mathcal{E}(u, \psi_x^{(m)}) = \int_K f \psi_x^{(m)} d\mu$$

as desired. $\square$

Lemma 2.14: Let $v: V^* \rightarrow \mathbb{R}$ be a harmonic function on the Vicsek set. If $v(x) = v(y_i) = 0$ for all $y_i \in V_{m,x}$ then $v(\omega) = 0$ for all $\omega \in \overline{T_x}$ where $T_x = \left(\bigcup_{k>0} S^k(x)\right) \cup \left(\bigcup_i \{y_i\}\right)$.

Proof. It is clear that $v(\omega) = 0$ for any $\omega \in T_x$, since the value of the points $S^k(x)$ with $k > 0$ depends on the value of x and its neighbours.

So we need to check the points in $\overline{T_x} \setminus T_x$. Now since T_x is dense in $\overline{T_x}$, then for any $\omega \in \overline{T_x}$ we have that there exists a sequence $x_n \in T_x$ such that $x_n \rightarrow \omega$. By the fact that v is continuous on K it is continuous on $\overline{T_x}$, in which case $|v(x_n) - v(\omega)| \rightarrow 0$ as $n \rightarrow \infty$, but $v(x_n) = 0$ for all x_n , so $v(\omega) = 0$ as desired. $\square$

What Lemma 2.14 states is, if any two points in V_m are neighbours, say x and y , and $v(x) = v(y) = 0$, then the function value at any point between x and y is also zero.

Lemma 2.15: Set $F_m = \text{supp } \psi_x^{(m)}$, then

$$F_0 \supset F_1 \supset F_2 \supset F_3 \supset \dots$$

Further F_m is contained in the open ball $B(x, \varepsilon_m)$ where $\varepsilon_m \rightarrow 0$ as $m \rightarrow \infty$.

Proof. If we set $\Omega_m = \left\{z \in K : \psi_x^{(m)}(z) = \psi_x^{(m)}(y) = 0, \forall y \in V_{m,z}\right\}$ then $F_m = K \setminus \left(\bigcup_{\alpha \in \Omega_m} \overline{B_\alpha}\right)$.

It is clear that $\Omega_m \subset \Omega_{m+1}$ and that there are new points in Ω_{m+1} that are not just a point between some already existing points $x_1, x_2 \in \Omega_m$. We therefore have, given the construction of F_m , that $F_m \supset F_{m+1}$.

The containment of F_m in $B(x, \varepsilon_m)$ follows from Lemma 2.14 and that $F_m \supset F_{m+1}$. $\square$

We are now ready to prove the pointwise formula.

Theorem 2.16: Let $u \in \text{dom } \Delta_\mu$. Then the pointwise formula

$$\Delta_\mu u(x) = \lim_{m \rightarrow \infty} r^{-m} \left(\int_K \psi_x^{(m)} d\mu \right)^{-1} \tilde{\Delta}_m u(x) \quad (2.4)$$

holds with the limit uniform across $V_* \setminus V_0$. Conversely, suppose u is a continuous function and the right side of (2.4) converges uniformly to a continuous function on $V_* \setminus V_0$. Then $u \in \text{dom } \Delta_\mu$ and (2.4) holds.

Proof. Suppose $u \in \text{dom } \Delta_\mu$ and $\Delta_\mu u = f$, then by Proposition 2.13 we have that

$$r^{-m} \left(\int_K \psi_x^{(m)} d\mu \right)^{-1} \tilde{\Delta}_m u(x) = \frac{\int_K f \psi_x^{(m)} d\mu}{\int_K \psi_x^{(m)} d\mu}$$

for any $x \in V_m \setminus V_0$. The right side needs to converge uniformly, so let $\varepsilon > 0$. Since $f(x)$ is continuous we can by Lemma 2.15 choose an m large enough such that for all $x, y \in F_m$ we have that $|f(x) - f(y)| < \varepsilon$, in which case

$$\begin{aligned} \left| \frac{\int_K f \psi_x^{(m)} d\mu}{\int_K \psi_x^{(m)} d\mu} - f(x) \right| &= \left| \frac{\int_K f \psi_x^{(m)} d\mu}{\int_K \psi_x^{(m)} d\mu} - \frac{\int_K f(x) \psi_x^{(m)} d\mu}{\int_K \psi_x^{(m)} d\mu} \right| \\ &= \left| \frac{\int_K (f(y) - f(x)) \psi_x^{(m)} d\mu}{\int_K \psi_x^{(m)} d\mu} \right| \\ &\leq \frac{\int_{F_m} |f(y) - f(x)| d\mu}{\int_K \psi_x^{(m)} d\mu} \\ &\leq \frac{\varepsilon \int_{F_m} d\mu}{\int_K \psi_x^{(m)} d\mu} \\ &= \varepsilon, \end{aligned}$$

therefore

$$\lim_{m \rightarrow \infty} r^{-m} \left(\int_K \psi_x^{(m)} d\mu \right)^{-1} \tilde{\Delta}_m u(x) = \lim_{m \rightarrow \infty} \frac{\int_K f \psi_x^{(m)} d\mu}{\int_K \psi_x^{(m)} d\mu} = f(x) = \Delta_\mu u(x)$$

proving the first statement.

Next suppose that the expression $r^m \left(\int_K \psi_x^{(m)} d\mu \right)^{-1} \tilde{\Delta}_m u(x)$ converges uniformly to a continuous function f . Then for any $v \in \text{dom}_0 \mathcal{E}$ we have

$$\mathcal{E}_m(u, v) = r^{-m} \sum_{\substack{x \sim y \\ m}} (u(y) - u(x)) (v(y) - v(x)).$$

If we collect all the terms that contain the factor $v(x)$ for a fixed $x \in V_m \setminus V_0$ then each $v(x)$ will be of the form $-v(x) (u(y) - u(x))$ for each $y \underset{m}{\sim} x$. We therefore get

$$\begin{aligned} \mathcal{E}_m(u, v) &= -r^{-m} \sum_{x \in V_m \setminus V_0} v(x) \left(\sum_{\substack{x \sim y \\ m}} (u(y) - u(x)) \right) \\ &= - \sum_{x \in V_m \setminus V_0} v(x) r^{-m} \tilde{\Delta}_m u(x). \end{aligned}$$

While it seems we lose many terms by removing $v(y)$, they are instead contained in

$$\sum_{\substack{x \sim y \\ m}} (u(y) - u(x)).$$

Note that the sum is "reset" for every x so we count both the relation $x \sim y$ and $y \sim x$.

Now if we define $f_m(x) = r^{-m} \left(\int_K \psi_x^{(m)}(y) d\mu \right)^{-1} \tilde{\Delta}_m u(x)$ then $f_m \rightarrow f$ uniformly, as $m \rightarrow \infty$, by assumption and we can rewrite the above as

$$\begin{aligned} \mathcal{E}_m(u, v) &= - \sum_{x \in V_m \setminus V_0} v(x) \left(\int_K \psi_x^{(m)}(y) d\mu \right) r^{-m} \left(\int_K \psi_x^{(m)}(y) d\mu \right)^{-1} \tilde{\Delta}_m u(x) \\ &= - \int_K \left(\sum_{x \in V_m \setminus V_0} v(x) f_m(x) \psi_x^{(m)}(y) \right) d\mu. \end{aligned}$$

Since the convergence, in the next calculation, follows from a quite lengthy argument, we will show the calculation and then return to the convergence. We have that

$$\begin{aligned} \mathcal{E}(u, v) &= \lim_{m \rightarrow \infty} \mathcal{E}_m(u, v) \\ &= \lim_{m \rightarrow \infty} - \int_K \left(\sum_{x \in V_m \setminus V_0} v(x) f_m(x) \psi_x^{(m)}(y) \right) d\mu = - \int_K v(y) f(y) d\mu \quad (2.5) \end{aligned}$$

therefore $u \in \text{dom} \Delta_\mu$. Now looking at the convergence in (2.5) we find

$$\begin{aligned} &\left| \sum_{x \in V_m \setminus V_0} v(x) f_m(x) \psi_x^{(m)}(y) - v(y) f(y) \right| \\ &\leq \left| \sum_{x \in V_m \setminus V_0} (v(x) f_m(x) - v(y) f(y)) \psi_x^{(m)}(y) \right| + \left| \sum_{x \in V_m \setminus V_0} v(y) f(y) \psi_x^{(m)}(y) - v(y) f(y) \right|. \end{aligned}$$

Looking at each term individually we first see that

$$\begin{aligned} &\left| \sum_{x \in V_m \setminus V_0} (v(x) f_m(x) - v(y) f(y)) \psi_x^{(m)}(y) \right| \\ &\leq \sum_{x \in V_m \setminus V_0} \left(|v(x) f_m(x) - v(y) f_m(y)| + |v(y) f_m(y) - v(y) f(y)| \right) \psi_x^{(m)}(y). \end{aligned}$$

We will again look at each term starting with $|v(y) f_m(y) - v(y) f(y)|$. Here

$$|v(y) f_m(y) - v(y) f(y)| = |v(y)| |f_m(y) - f(y)|$$

and since v is continuous on K , which is compact, there is a constant C_1 so $|v(y)| \leq C_1$ for all $x \in K$. Because f_m converges uniformly to f , we can choose m_1 such that $|f_{m_1}(y) - f(y)| \leq \frac{\varepsilon}{C_1}$ and the convergence of the term follows.

Next is $|v(x) f_m(x) - v(y) f_m(y)|$. Here we find

$$|v(x) f_m(x) - v(y) f_m(y)| \leq |v(x)| |f_m(x) - f(y)| + |f_m(y)| |v(x) - v(y)|.$$

First we look at $|f_m(y)||v(x) - v(y)|$. Here $|f_m(y)| \leq C_2$ and by Lemma 2.15 and continuity of v we can choose m_2 so $|v(x) - v(y)| \leq \frac{\varepsilon}{C_2}$ for all $x \in F_{m_2}$. The reason we can choose m_2 is because $\psi_x^{(m)}$ is multiplied onto all the terms restricting v to F_{m_2} .

Second we examine $|v(x)||f_m(x) - f(y)|$. Again $|v(x)| \leq C_1$, but

$$|f_m(x) - f(y)| \leq |f_m(x) - f(x)| + |f(x) - f_m(y)| \leq |f_m(x) - f(x)| + |f(x) - f(y)| + |f(y) - f_m(y)|$$

First choose m_3 so $|f_{m_3}(x) - f(x)| \leq \frac{\varepsilon}{3C_1}$. Next by continuity of f choose m_4 so $|f(x) - f(y)| \leq \frac{\varepsilon}{3C_1}$ for all $x \in F_{m_4}$. Lastly choose m_5 so $|f(y) - f_{m_5}(y)| \leq \frac{\varepsilon}{3C_1}$.

So we now have for $m \geq \max\{m_1, m_2, \dots, m_5\}$ that

$$\left| \sum_{x \in V_m \setminus V_0} (v(x)f_m(x) - v(y)f(y)) \psi_x^{(m)}(y) \right| \leq \varepsilon \sum_{x \in V_m \setminus V_0} \psi_x^{(m)}(y) \leq \varepsilon.$$

We are going to look at

$$\left| \sum_{x \in V_m \setminus V_0} v(y)f(y)\psi_x^{(m)}(y) - v(y)f(y) \right| = |v(y)f(y)| \left| \sum_{x \in V_m \setminus V_0} \psi_x^{(m)}(y) - 1 \right|.$$

and show that

$$\left| \sum_{x \in V_m \setminus V_0} \psi_x^{(m)}(y) - 1 \right| = 0, \quad \text{as } m \rightarrow \infty. \quad (2.6)$$

If $y \in V^*$, then choose m_6 so $y \in V_{m_6}$ and (2.6) holds.

Next assume $y \in K \setminus V^*$, then since V^* is dense in K we can choose a sequence $(x_n)_{n \geq 1}$ where $x_n \rightarrow y$ as $n \rightarrow \infty$, and since $\psi_x^{(m)}$ is continuous on K then $\psi_x^{(m)}(x_n) \rightarrow \psi_x^{(m)}(y)$ as $n \rightarrow \infty$. This gives us the following

$$\left| \sum_{x \in V_m \setminus V_0} \psi_x^{(m)}(y) - 1 \right| \leq \left| \sum_{x \in V_m \setminus V_0} \psi_x^{(m)}(y) - \sum_{x \in V_m \setminus V_0} \psi_x^{(m)}(x_n) \right| + \left| \sum_{x \in V_m \setminus V_0} \psi_x^{(m)}(x_n) - 1 \right|.$$

Now choose an n_1 so $|\psi_x^{(m)}(x_{n_1}) - \psi_x^{(m)}(y)| \leq \varepsilon$ and an m_7 so $x_{n_1} \in V_{m_7}$. Both can then be bounded by $m \geq \max\{n_1, m_7\}$. Thus we have the convergence of (2.5). $\square$

Lastly we calculate the constants $\left(\int_K \psi_x^{(m)}(y) d\mu(y) \right)^{-1}$.

Lemma 2.17:

$$\int_K \psi_x^m(y) d\mu(y) = \begin{cases} 5^{-m} & \text{if } \#V_{m,p} = 4 \\ 2 \cdot 5^{-(m+1)} & \text{if } \#V_{m,p} = 2 \\ 5^{-(m+1)} & \text{if } \#V_{m,p} = 1. \end{cases}$$

Proof. Recall from eq. (1.3) that we can write a point x 's "address" with the infinite sequence $K_{i_1, i_2, \dots} = (\mathcal{S}_{i_1} \circ \mathcal{S}_{i_2} \circ \dots)(K)$. These sequences are not necessarily unique and we will have three cases. The first two cases are where x has a unique sequence and $\#V_{m,p} \in \{1, 4\}$. The last case is where x has two sequences and $\#V_{m,p} = 2$. If we start with the two sequences, then we have that $x \in K_{i_1, \dots, i_m} \cap K_{i_1, \dots, i_{m'}}$ in which case from Lemma 2.14 it follows that $\text{supp } \psi_x^{(m)} = K_{i_1, \dots, i_{m+1}} \cup K_{i_1, \dots, i_{m'+1}}$, for an appropriate choice of i_{m+1} and $i_{m'+1}$. Now if $x \in K_{i_1, \dots, i_m}$ and $\#V_{m,p} = 4$ we have that $\text{supp } \psi_x^{(m)} = K_{i_1, \dots, i_m}$, but if $\#V_{m,p} = 1$ then $\text{supp } \psi_x^{(m)} = K_{i_1, \dots, i_{m+1}}$ again for an appropriate i_{m+1} .

If we now look at an $\alpha \in \text{supp } \psi_x^{(m)}$ it is going to have a sequence of the form $K_{i_1, \dots, i_m, \alpha_{j_1}, \alpha_{j_2} \dots}$ or $K_{i_1, \dots, i_k, \alpha_{j_1}, \alpha_{j_2} \dots}$ where $k \in \{m+1, m'+1\}$. We therefore have

$$\int_K \psi_x^{(m)}(y) \, d\mu(y) = \int_{\text{supp } \psi_x^{(m)}} \psi_x^{(m)}(y) \, d\mu(y) = \sum_{\alpha \in \text{supp } \psi_x^{(m)}} \int_{\{\alpha\}} \psi_x^{(m)}(y) \, d\mu(y).$$

If x has a unique sequence and $\#V_{m,p} = 4$ then

$$\begin{aligned} \sum_{\alpha \in \text{supp } \psi_x^{(m)}} \int_{\{\alpha\}} \psi_x^{(m)}(y) \, d\mu(y) &= \sum_{\alpha \in \text{supp } \psi_x^{(m)}} \mu((\mathcal{S}_{i_1} \circ \dots \circ \mathcal{S}_{i_m} \circ \mathcal{S}_{\alpha_{j_1}} \circ \mathcal{S}_{\alpha_{j_2}} \dots)(K)) \\ &= \mu\left(\bigcup_{(j_1, j_2, \dots) \in \mathcal{I}} (\mathcal{S}_{i_1} \circ \dots \circ \mathcal{S}_{i_m} \circ \mathcal{S}_{\alpha_{j_1}} \circ \mathcal{S}_{\alpha_{j_2}} \circ \dots)(K)\right) = \mu((\mathcal{S}_{i_1} \circ \dots \circ \mathcal{S}_{i_m})(K)) = 5^{-m}. \end{aligned}$$

By the almost exact same calculation we find that if x has a unique sequence and $\#V_{m,p} = 1$ then

$$\sum_{\alpha \in \text{supp } \psi_x^{(m)}} \int_{\{\alpha\}} \psi_x^{(m)}(y) \, d\mu(y) = 5^{-(m+1)}.$$

Now if x does not have a unique sequence then

$$\begin{aligned} \sum_{\alpha \in \text{supp } \psi_x^{(m)}} \int_{\{\alpha\}} \psi_x^{(m)}(y) \, d\mu(y) &= \sum_{\alpha \in \text{supp } \psi_x^{(m)}} \left[\mu((\mathcal{S}_{i_1} \circ \dots \circ \mathcal{S}_{i_{m+1}} \circ \mathcal{S}_{\alpha_{j_1}} \circ \mathcal{S}_{\alpha_{j_2}} \dots)(K)) \right. \\ &\quad \left. + \mu((\mathcal{S}_{i_1} \circ \dots \circ \mathcal{S}_{i'_m+1} \circ \mathcal{S}_{\alpha_{j_1}} \circ \mathcal{S}_{\alpha_{j_2}} \dots)(K)) \right] \\ &= \mu\left(\bigcup_{(j_1, j_2, \dots) \in \mathcal{I}} (\mathcal{S}_{i_1} \circ \dots \circ \mathcal{S}_{i_{m+1}} \circ \mathcal{S}_{\alpha_{j_1}} \circ \mathcal{S}_{\alpha_{j_2}} \circ \dots)(K)\right) \\ &\quad + \mu\left(\bigcup_{(j_1, j_2, \dots) \in \mathcal{I}} (\mathcal{S}_{i_1} \circ \dots \circ \mathcal{S}_{i'_m+1} \circ \mathcal{S}_{\alpha_{j_1}} \circ \mathcal{S}_{\alpha_{j_2}} \circ \dots)(K)\right) \\ &= \mu((\mathcal{S}_{i_1} \circ \dots \circ \mathcal{S}_{i_{m+1}})(K)) + \mu((\mathcal{S}_{i_1} \circ \dots \circ \mathcal{S}_{i'_m+1})(K)) \\ &= 2 \cdot 5^{-(m+1)} \end{aligned}$$

and the result follows. $\square$

2.3 The Dirichlet Laplacian

We will now construct the Dirichlet Laplacian. Since the Dirichlet Laplacian is a densely defined operator we will therefore start by recalling some definitions for a densely defined operator on a Hilbert space. We will also need Dirichlet forms, thus we will also be defining them.

Definition 2.18: Let $\mathcal{H}$ be a Hilbert space and let $A: \text{dom } A \rightarrow \mathcal{H}$ be a densely defined operator. Then we say the following about A :

- (i) A is said to be symmetric if for $f, g \in \text{dom } A$

$$\langle Af, g \rangle = \langle f, Ag \rangle.$$

- (ii) A is said to be non-negative (non-positive resp.) if for $f \in \text{dom } A$

$$\langle Af, f \rangle \geq 0, \quad (\langle Af, f \rangle \leq 0 \text{ resp.}).$$

- (iii) A is said to be self-adjoint if $\text{dom } A^* = \text{dom } A$ where A^* is the adjoint of A on the domain

$$\text{dom } A^* = \left\{ f \in \mathcal{H} : \exists c(f) \geq 0, \forall g \in \mathcal{D}(A), |\langle f, Ag \rangle| \leq c(f) \|g\| \right\}$$

where $c(f)$ denotes a constant that may depend on f . We can further define A^* by the formula

$$\langle A^* f, g \rangle = \langle f, Ag \rangle, \quad \text{for } f \in \text{dom } A^*, g \in \text{dom } A.$$

Definition 2.19: A quadratic form $\mathcal{E}$ on $\mathcal{H}$ is a non-negative, definite, symmetric bilinear form from $\text{dom } \mathcal{E} \times \text{dom } \mathcal{E} \rightarrow \mathbb{R}$, where $\text{dom } \mathcal{E}$ is a dense subspace of $\mathcal{H}$, where

- (i) A quadratic form is said to be closed if $\text{dom } \mathcal{E}$ equipped with the norm

$$\|f\|_{\text{dom } \mathcal{E}}^2 = \|f\|^2 + \mathcal{E}(f, f)$$

is a Hilbert space.

- (ii) A quadratic form $\mathcal{E}$ on $\mathcal{H}$ is said to be closable if it admits a closed extension. That is, there exists a closed quadratic form $\bar{\mathcal{E}}$ such $\text{dom } \mathcal{E} \subset \text{dom } \bar{\mathcal{E}}$ and $\bar{\mathcal{E}}$ coincides with $\mathcal{E}$ on $\text{dom } \mathcal{E} \times \text{dom } \mathcal{E}$.

- (iii) We further call a quadratic form $\mathcal{E}$ a Dirichlet form if it has the following property. If $u, v \in \text{dom } \mathcal{E}$ so that for almost every $x, y \in K$ we have

$$|v(x) - v(y)| \leq |u(x) - u(y)|, \quad \text{and} \quad |v(x)| \leq |u(x)|$$

then

$$\mathcal{E}(v, v) \leq \mathcal{E}(u, u).$$

Now recall from Definition 2.8 that

$$\mathcal{E}(f, f) = \lim_{m \rightarrow \infty} \mathcal{E}_m(f, f)$$

with domain $\text{dom}_0 \mathcal{E}$ where

$$\mathcal{E}_m(f, f) = \sum_{\substack{x \sim y \\ m}} (f(x) - f(y))^2.$$

By [8, Theorem 4.1] we have that $(\mathcal{E}, \text{dom}_0 \mathcal{E})$ is a Dirichlet form on $L^2(K, \mu)$. By [33, Lemma 1.3.3] we have that

$$\mathcal{E}(f, f) = - \int_K \Delta f g \, d\mu,$$

where Δ a unique densely defined self-adjoint operator. The operator Δ is called the generator of $\mathcal{E}$. Moreover we have that Δ is an extension of the Kigami Laplacian Δ_μ . The operator Δ with domain $\text{dom } \Delta$ is referred to as the Dirichlet Laplacian.

Notice that the Kigami Laplacian Δ_μ was not used in the construction of the Dirichlet Laplacian Δ , but that we used the limit of the renormalized graph energy from Definition 2.8. This stems from the fact that there are two ways to define Δ , one through Dirichlet forms and one through Δ_μ . We have chosen to define Δ through Dirichlet forms, but defining Δ through Δ_μ follows a similarly condensed argument. One would prove that $\Delta_\mu: \text{dom } \Delta_\mu \rightarrow L^2(K, \mu)$ is a densely defined non-positive symmetric operator. It then follows from [33, Lemma 1.3.5] that the quadratic form $\mathcal{E}(f, g) = -\langle f, \Delta_\mu g \rangle$ for $f, g \in \text{dom } \Delta_\mu$ is closable and the generator of $\mathcal{E}$ is the self-adjoint extension Δ .

2.4 The Discrete Fractional Laplacian

Recalling Definition 2.2 we consider the eigenvalues $0 < \lambda_1^m \leq \lambda_2^m \leq \dots \leq \lambda_{N_m}^m$, each being repeated accordingly to multiplicity, of $-\Delta_m$. Next let $(\phi_i^m)_{1 \leq i \leq N_m}$ be the corresponding eigenfunctions with respect to the measure μ_m . This leads us to defining the following.

Definition 2.20: Let $s \geq 0$, we then define the *discrete Riesz kernel* on V_m as

$$G_s^m(x, y) = \sum_{i=1}^{N_m} (\lambda_i^m)^{-s} \phi_i^m(x) \phi_i^m(y), \quad \text{for } x, y \in V_m.$$

By construction we can see that the matrix $(G_s^m(x, y))_{x, y \in V_m}$ is symmetric. It is also positive semidefinite, which follows from

$$\begin{aligned} \sum_{i,j=1}^{N_m} a_i a_j G_s^m(x_i, x_j) &= \sum_{i,j=1}^{N_m} a_i a_j \sum_{k=1}^{N_m} (\lambda_k^m)^{-s} \phi_k^m(x_i) \phi_k^m(x_j) \\ &= \sum_{k=1}^{N_m} (\lambda_k^m)^{-s} \left(\sum_{i=1}^{N_m} \phi_k(x_i)^2 a_i^2 + \sum_{i,j=1, i \neq j}^{N_m} \phi_k(x_i) \phi_k(x_j) a_i a_j \right) \\ &= \sum_{k=1}^{N_m} (\lambda_k^m)^{-s} \left(\sum_{i=1}^{N_m} \phi_k(x_i) a_i \right)^2 \geq 0. \end{aligned}$$

Thus it can be the covariance matrix of a Gaussian vector. The Riesz kernel will allow us to define the discrete fractional Laplacian.

Definition 2.21: For $f \in \ell(V_m)$ and $s \geq 0$, the *discrete fractional Laplacian* $(-\Delta_m)^s: L^2(V_m, \mu_m) \rightarrow L^2(V_m, \mu_m)$ is defined by

$$(-\Delta_m)^{-s} f(x) = \frac{1}{a_m} \sum_{y \in V_m} G_s^m(x, y) f(y) = \sum_{i=1}^{N_m} (\lambda_i^m)^{-s} \phi_i^m(x) \frac{1}{a_m} \sum_{y \in V_m} \phi_i^m(y) f(y).$$

Remark 2.22: Note that $\frac{1}{a_m} \sum_{y \in V_m} \phi_i^m(y) f(y) = \langle \phi_i^m, f \rangle_{L^2(V_m, \mu_m)}$ and therefore that $(-\Delta_m)^{-s} f(x) = \sum_{i=1}^{N_m} (\lambda_i^m)^{-s} \phi_i^m(x) \langle \phi_i^m, f \rangle_{L^2(V_m, \mu_m)}$.

Having stated the discrete fractional Laplacian we move on to proving some of its useful properties.

Proposition 2.23: Let $(-\Delta_m)^{-s}$ be defined as in Definition 2.21 then the following holds

- (i) $(-\Delta_m)^{-s}$ is a uniformly bounded operator.
- (ii) $(-\Delta_m)^{-s}$ is a symmetric operator.

Proof. Here $\|\cdot\| = \|\cdot\|_{L^2(V_m, \mu_m)}$ and $\langle \cdot, \cdot \rangle = \langle \cdot, \cdot \rangle_{L^2(V_m, \mu_m)}$. We start with (i). We have

$$\begin{aligned} \|(-\Delta_m)^{-s} f(x)\|^2 &\leq \left\| \sum_{i=1}^{N_m} (\lambda_i^m)^{-s} \phi_i^m(x) \frac{1}{a_m} \sum_{y \in V_m} \phi_i^m(y) f(y) \right\|^2 \\ &\leq \sum_{i=1}^{N_m} (\lambda_i^m)^{-2s} \left\| \phi_i^m(x) \frac{1}{a_m} \sum_{y \in V_m} \phi_i^m(y) f(y) \right\|^2 \\ &\leq (\lambda_1^m)^{-2s} \sum_{i=1}^{N_m} |\langle \phi_i^m, f \rangle|^2 \\ &\leq (\lambda_1^m)^{-2s} \|f\|^2, \end{aligned}$$

where the second to last inequality comes from the fact that $0 < \lambda_1 < \lambda_2 < \dots$. The uniform bound comes from the fact that $\inf_m \lambda_1^m > 0$. This follows from [8, Lemma 5.2], since if $\inf_m \lambda_1^m \leq 0$ then the series in the Lemma would not converge.

Moving on to (ii), we have that

$$\begin{aligned}
\langle f, (-\Delta_m)^{-s} g \rangle &= \int_{V_m} f(-\Delta_m)^{-s} g \, d\mu_m(x) \\
&= \int_{V_m} f(x) \left(\sum_{j=1}^{N_m} (\lambda_j^m)^{-s} \phi_j^m(x) \frac{1}{a_m} \sum_{y \in V_m} \phi_j^m(y) g(y) \right) d\mu_m(x) \\
&= \sum_{j=1}^{N_m} (\lambda_j^m)^{-s} \frac{1}{a_m} \left(\sum_{y \in V_m} \phi_j^m(y) g(y) \right) \int_{V_m} \left(\sum_{i=1}^{N_m} \phi_i^m(x) \langle \phi_i^m, f \rangle \right) \phi_j^m(x) d\mu_m(x) \\
&= \sum_{j=1}^{N_m} (\lambda_j^m)^{-s} \frac{1}{a_m} \left(\sum_{y \in V_m} \phi_j^m(y) g(y) \right) \int_{V_m} \phi_j^m(x) \langle \phi_j^m, f \rangle \phi_j^m(x) d\mu_m(x) \\
&= \sum_{j=1}^{N_m} \left[\int_{V_m} \left((\lambda_j^m)^{-s} \phi_j(x) \langle \phi_j^m, f \rangle \right) \phi_j^m(x) \langle \phi_j^m, g \rangle d\mu_m(x) \right. \\
&\quad \left. + \int_{V_m} \left((\lambda_j^m)^{-s} \phi_j^m(x) \langle \phi_j^m, f \rangle \right) \left(\sum_{i=1, i \neq j}^{N_m} \phi_i^m(x) \langle \phi_i^m, g \rangle \right) d\mu_m(x) \right] \\
&= \int_{V_m} (-\Delta_m)^{-s} f g \, d\mu_m(x) \\
&= \langle (-\Delta_m)^{-s} f, g \rangle
\end{aligned}$$

using that $f = \sum_{i=1}^{N_m} \phi_i \langle \phi_i, f \rangle$ and $\int_{V_m} \phi_i \phi_j \, d\mu_m$ is 1 if $j = i$ and 0 if $i \neq j$. $\square$

2.5 Semigroups and Heat kernels

When constructing the fractional Laplacian in Section 2.6, we will use that $-\Delta$ generates a semigroup which admits a heat kernel. We will therefore first define a semigroup as well as showing that $-\Delta$ generates a semigroup. We will also define the heat kernel.

Definition 2.24: Let $\mathcal{H}$ be a Hilbert space. A *strongly continuous self-adjoint contraction semigroup* is a family of self-adjoint operators $(P_t)_{t \geq 0}: \mathcal{H} \rightarrow \mathcal{H}$ which satisfies the following properties:

- (i) For $s, t \geq 0$, $P_t \circ P_s = P_{s+t}$ (semigroup property).
- (ii) For every $f \in \mathcal{H}$, $\lim_{t \rightarrow 0} P_t f = f$ (strong continuity).
- (iii) For every $f \in \mathcal{H}$ and $t \geq 0$, $\|P_t f\| \leq \|f\|$ (contraction property).

The following theorem is vital as it will show that an operator can generate a semigroup.

Theorem 2.25: If A is a densely defined self adjoint operator on $\mathcal{H}$ then it is the generator of the strongly continuous self-adjoint contraction semigroup on $\mathcal{H}$ defined as $P_t = e^{-tA}$.

Proof. Let A be given as stated. From the spectral theorem [33, Theorem 1.1.2] there is a measure space (Ω, ν) , a unitary map $U: L^2(\Omega, \nu) \rightarrow \mathcal{H}$ and a non negative real valued measurable function λ on Ω such that

$$U^{-1}AUf(x) = -\lambda(x)f(x)$$

for $x \in \Omega, Uf \in \text{dom } A$. Then from the functional calculus [33, Theorem 1.1.3] we define $P_t: \mathcal{H} \rightarrow \mathcal{H}$ such that

$$U^{-1}P_tUf(x) = e^{-t\lambda(x)}f(x).$$

Note that since U is a unitary map, it is a linear bijection that for all $x, y \in L^2(\Omega, \nu)$ satisfies $\langle Ux, Uy \rangle_{\mathcal{H}} = \langle x, y \rangle_{L^2(\Omega, \nu)}$.

It is enough to show that $Q_t = U^{-1}P_tU$ is self-adjoint, since then P_t will be self-adjoint. This can be proven quite easily. If we have the functions $f, g \in \mathcal{H}$ and $\psi, \phi \in L^2(\Omega, \nu)$ then P_t is symmetric since

$$\begin{aligned} \langle Af, g \rangle_{\mathcal{H}} &= \langle AU\psi, U\phi \rangle_{\mathcal{H}} = \langle UU^{-1}UAU\psi, UU^{-1}U\phi \rangle_{\mathcal{H}} = \langle U^{-1}AU\psi, \phi \rangle_{L^2(\Omega, \nu)} \\ &= \langle \psi, U^{-1}AU\phi \rangle_{L^2(\Omega, \nu)} = \langle U\psi, UU^{-1}AU\phi \rangle_{\mathcal{H}} = \langle f, Ag \rangle_{\mathcal{H}}. \end{aligned}$$

That $\text{dom } P_t^* = \text{dom } P_t$ reasons from the same argument as in (2.7), which will be given later in the proof.

We will prove that $Q_t = U^{-1}P_tU$ with domain $\text{dom } Q_t$ is a self-adjoint strongly continuous contraction semigroup on $L^2(\Omega, \nu)$ with generator A . Starting with A being a generator, we see that for $f \in \text{dom } A$

$$\begin{aligned} \lim_{t \rightarrow 0} \left\| \frac{Q_tf - f}{t} - Af \right\| &= \lim_{t \rightarrow 0} \sqrt{\left\langle \frac{Q_tf - f}{t} - Af, \frac{Q_tf - f}{t} - Af \right\rangle} \\ &= \lim_{t \rightarrow 0} \sqrt{\left\langle \frac{Q_tf - f}{t}, \frac{Q_tf - f}{t} - Af \right\rangle - \left\langle Af, \frac{Q_tf - f}{t} - Af \right\rangle} \\ &\leq \lim_{t \rightarrow 0} \sqrt{\left\langle \frac{Q_tf - f}{t}, \frac{Q_tf - f}{t} - Af \right\rangle} \\ &= \lim_{t \rightarrow 0} \sqrt{\int_{\Omega} \left(\frac{Q_tf - f}{t} \right) \left(\frac{Q_tf - f}{t} - Af \right) d\nu} \\ &= \lim_{t \rightarrow 0} \sqrt{\int_{\Omega} \left(\frac{(e^{-\lambda t} - 1)f}{t} \right) \left(\frac{Q_tf - f}{t} - Af \right) d\nu} \\ &= 0 \end{aligned}$$

where the convergence comes from the fact that $e^{-\lambda t}$ converges to 1 much faster than $1/t$ diverges to ∞ . Thus A is a generator. Next we have that Q_t is self-adjoint, for which we start with symmetry

$$\langle Q_tf, g \rangle = \int_{\Omega} Q_tf g d\nu = \int_{\Omega} e^{-\lambda t} f g d\nu = \langle f, Q_tg \rangle.$$

Furthermore $\text{dom } Q_t$ is dense in $L^2(\Omega, \nu)$ since if $f_n \rightarrow f$ in $L^2(\Omega, \nu)$ and $Q_t f_n \rightarrow g$ then

$$\lim_{n \rightarrow \infty} Q_t f_n = \lim_{n \rightarrow \infty} e^{-\lambda t} f_n = e^{-\lambda t} f = Q_t f.$$

Showing that $\text{dom } Q_t^* = \text{dom } Q_t$. Since Q_t is symmetric we have from the Cauchy-Schwartz inequality that

$$|\langle f, Q_t g \rangle| = |\langle Q_t f, g \rangle| \leq \|Q_t f\| \|g\|. \quad (2.7)$$

Thus we have Q_t is self-adjoint.

To show the semigroup properties, we have Definition 2.24.(i)

$$Q_{t+s} f = e^{-\lambda(t+s)} f = e^{-\lambda t} e^{-\lambda s} f = Q_t Q_s f.$$

Further for Definition 2.24.(ii) we have

$$\lim_{t \rightarrow \infty} Q_t f = \lim_{t \rightarrow \infty} e^{-\lambda t} f = f$$

and lastly Definition 2.24.(iii)

$$\|Q_t f\|^2 = \int_{\Omega} (e^{-\lambda t} f)^2 d\nu \leq \int_{\Omega} f^2 d\nu = \|f\|^2.$$

Thus Q_t is a strongly continuous self-adjoint contraction semigroup on $L^2(\Omega, \nu)$ with generator A . $\square$

The next corollary follows quite easily from Theorem 2.25.

Corollary 2.26: P_t is a linear semigroup.

Proof. Let $f, g \in \mathcal{H}$ and α be some constant. Then

$$\begin{aligned} U^{-1} P_t U (\alpha f(x) + g(x)) &= e^{-t\lambda(x)} (\alpha f(x) + g(x)) \\ &= \alpha e^{-t\lambda(x)} f(x) + e^{-t\lambda(x)} g(x) = \alpha U^{-1} P_t U f(x) + U^{-1} P_t U g(x) \end{aligned}$$

$\square$

With semigroups out of the way we can define the heat kernel.

Definition 2.27: Let $(P_t)_{t \geq 0}$ be a strongly continuous self-adjoint contraction semigroup on $L^2(X, \mu)$. We say that the semigroup $(P_t)_{t \geq 0}$ admits a heat kernel if the heat kernel measure has a density with respect to μ , i.e. there exists a measurable function $p: \mathbb{R}_{>0} \times X \times X \rightarrow \mathbb{R}_{\geq 0}$ such that for every $t > 0$, a.e. we have

$$P_t f(x) = \int_K p_t(x, y) f(y) d\mu(y)$$

with $x, y \in X, f \in L^\infty(X, \mu)$.

Remark 2.28: The discrete Laplacian $-\Delta_m$ also generates a semigroup. This is because Δ_m can be represented as the matrix L_m from Definition 2.2. Since L_m is symmetric, it can be diagonalized in an orthonormal basis $(\Phi_i)_{1 \leq i \leq N_m}$, where $(\Phi_i)_j = \phi_i(x_j)$ for $x_i \in V_m$. We then have for every vector $X \in \mathbb{R}^{N_m}$ that

$$e^{-tL_m}X = \sum_{i=1}^{N_m} e^{-t\lambda_i} \langle X, \Phi_i \rangle \Phi_i.$$

It is obvious to see that $P_t^m = e^{-tL_m}$ is a semigroup, as the reasoning follows the same given in Theorem 2.25. As we will want to use this semigroup on functions from $\ell(V_m)$ we will, for an $f \in \ell(V_m)$ and $x_j \in V_m$ write

$$e^{-tL_m}f(x_j) = \sum_{i=1}^{N_m} e^{-t\lambda_i} \langle f, \Phi_i \rangle_{L^2(V_m, \mu_m)} (\Phi_i)_j$$

2.6 The Fractional Laplacian

Before we can define the continuous fractional Laplacian, we need to define the eigenvalues and eigenfunctions for $-\Delta$.

Recall that the Laplacian Δ , with domain $\text{dom } \Delta$, is a densely defined self-adjoint operator. Therefore by Theorem 2.25 it is the generator of a strongly continuous Markov semigroup $\{P_t\}_{t \geq 0}$ on $L^2(K, \mu)$.

From [10, Theorem 8.4] this semigroup admits a heat kernel $p_t(x, y)$, $t > 0, x, y \in K$, with respect to the normalized Hausdorff measure μ . The heat kernel $p_t(x, y)$ is called the Dirichlet heat kernel on K . Furthermore we get that from [23, section 2.3] this heat kernel satisfies, for some $c_1, c_2 \in (0, \infty)$,

$$p_t(x, y) \leq c_1 t^{-d_h/d_w} \exp \left(-c_2 \left(\frac{d(x, y)^{d_w}}{t} \right)^{\frac{1}{d_w-1}} \right) \quad (2.8)$$

for every $x, y \in K \times K$ and $t \in (0, \infty)$ where d_h is the Hausdorff dimension and d_w is the walk dimension.

The Dirichlet heat kernel is quite central in the definition of the continuous fractional Laplacian, as its bounds and definition will be used quite a bit throughout the proofs in the following sections. One of the more important properties of the heat kernel is, that it can be described as a sum of eigenvalues and eigenfunctions of Δ . The following theorem gives us the existence of the eigenvalues, functions and the sum representation of the Dirichlet heat kernel. This theorem will not be proved, but a detailed proof can be found in [4, Theorem 2.2].

Theorem 2.29: There exists an orthonormal basis $\{\phi_i\}_{i=1}^\infty$ of $L^2(K, \mu)$ with ϕ_j being eigenfunctions of $-\Delta$. If λ_i is the eigenvalue belonging to ϕ_i , then $\lambda_i > 0$ and $\lim_{i \rightarrow \infty} \lambda_i = \infty$. Moreover

$$p_t(x, y) = \sum_{j=1}^{\infty} e^{-\lambda_j t} \phi_j(x) \phi_j(y),$$

where the convergence is uniform for all $t \in (0, \infty)$ and $x, y \in K \times K$.

The eigenvalues are not only used to represent the heat kernel but will also be used to represent the fractional Laplacian. The following lemma gives us an idea of what parameter value is need for convergence.

Lemma 2.30: Let $0 < \lambda_1 \leq \lambda_2 < \dots$ be the eigenvalues of $-\Delta$, then the following holds

$$\sum_{j=1}^{\infty} \frac{1}{\lambda_j^{2s}} < \infty \iff s > \frac{d_h}{2d_w}.$$

Proof. This follows from [12, Lemma 5.1.3] which states that there exists constants $c_1, c_2 > 0$ such that

$$c_1 \cdot n^{2/d_s} \leq \lambda_n \leq c_2 \cdot n^{2/d_s}. \quad (2.9)$$

Assume $s > \frac{d_h}{2d_w}$, then by (2.9)

$$\sum_{j=1}^{\infty} \frac{1}{\lambda_j^{2s}} \leq \sum_{j=1}^{\infty} \frac{1}{(c_1 n^{2/d_s})^{2s}} < \infty,$$

since

$$\frac{4s}{d_s} > \frac{2d_h}{2d_h/d_w} = 1.$$

Hence the series is convergent.

Assume now that

$$\sum_{j=1}^{\infty} \frac{1}{\lambda_j^{2s}} < \infty.$$

Then again by (2.9) we have that

$$\sum_{j=1}^{\infty} \frac{1}{(c_2 n^{2/d_s})^{2s}} \leq \sum_{j=1}^{\infty} \frac{1}{\lambda_j^{2s}} < \infty,$$

and the left sum only converges if $4s/d_s > 1$ in which case $s > d_s/4 = d_h/2d_w$ as desired. $\square$

Thus we are ready to define the fractional Laplacian.

Definition 2.31: Let $s \geq 0$, and $f \in L^2(K, \mu)$. The *fractional Laplacian* $(-\Delta)^{-s}$ is defined as

$$(-\Delta)^{-s} f = \sum_{j=1}^{\infty} \frac{1}{\lambda_j^s} \phi_j \int_K \phi_j(y) f(y) \, d\mu(y).$$

Remark 2.32: Once again we notice that $\langle \phi_i, f \rangle = \int_K \phi_i f \, d\mu$, and can rewrite

$$(-\Delta)^{-s} f = \sum_{j=1}^{\infty} \frac{1}{\lambda_j^s} \phi_j \langle \phi_j, f \rangle.$$

Now that we have the fractional Laplacian defined, we will show some of its very useful properties.

Proposition 2.33: Let $(-\Delta)^{-s}$ be defined as in Definition 2.31. Then:

- (i) $(-\Delta)^{-s}$ is a linear operator.
- (ii) If ϕ_n is an orthonormal eigenfunction, then $(-\Delta)^{-s}\phi_n = \frac{1}{\lambda_n^{-s}}\phi_n$.
- (iii) $(-\Delta)^{-s}$ is a bounded operator.
- (iv) $(-\Delta)^{-s}$ is a symmetric operator.

Proof. We begin with the proof of (i).

Let $f, g \in L^2(K, \mu)$ and let α be some constant. Then by definition of $(-\Delta)^{-s}$ we have

$$\begin{aligned} (-\Delta)^{-s}(\alpha f + g) &= \sum_{j=1}^{\infty} \frac{1}{\lambda_j^s} \phi_j \int_K \phi_j(y) (\alpha f(y) + g(y)) \, d\mu(y) \\ &= \alpha \sum_{j=1}^{\infty} \frac{1}{\lambda_j^s} \phi_j \int_K \phi_j(y) f(y) \, d\mu(y) + \sum_{j=1}^{\infty} \frac{1}{\lambda_j^s} \phi_j \int_K \phi_j(y) g(y) \, d\mu(y) \\ &= \alpha (-\Delta)^{-s} f + (-\Delta)^{-s} g. \end{aligned}$$

Moving on to (ii). We note that

$$\int_K \phi_j \phi_i \, d\mu = \delta_{ij},$$

where δ_{ij} is the Kronecker delta. We then have for any eigenfunction ϕ_n that

$$(-\Delta)^{-s}\phi_n = \sum_{j=1}^{\infty} \frac{1}{\lambda_j^s} \phi_j(x) \int_K \phi_j(y) \phi_n(y) \, d\mu(y) = \frac{1}{\lambda_n^{-s}} \phi_n.$$

Next we have (iii). This follows from

$$\begin{aligned} \|(-\Delta)^{-s}f\|^2 &= \left\| \sum_{j=1}^{\infty} \frac{1}{\lambda_j^s} \phi_j \langle \phi_j, f \rangle \right\|^2 \\ &= \left\langle \sum_{j=1}^{\infty} \frac{1}{\lambda_j^s} \phi_j \langle \phi_j, f \rangle, \sum_{j=1}^{\infty} \frac{1}{\lambda_j^s} \phi_j \langle \phi_j, f \rangle \right\rangle \\ &= \sum_{j=1}^{\infty} \left| \frac{1}{\lambda_j^s} \langle \phi_j, f \rangle \right|^2 \langle \phi_j, \phi_j \rangle \\ &\leq \left(\frac{1}{\lambda_1^s} \right)^2 \sum_{j=1}^{\infty} |\langle \phi_j, f \rangle|^2 \\ &= \left(\frac{1}{\lambda_1^s} \right)^2 \|f\|^2, \end{aligned}$$

in which case we have our bound. Finally we will show (iv)

$$\begin{aligned}
\langle (-\Delta)^{-s} f, g \rangle &= \int_K \left(\sum_{j=1}^{\infty} \frac{1}{\lambda_j^s} \phi_j \langle \phi_j, f \rangle \right) g \, d\mu \\
&= \int_K \left(\sum_{j=1}^{\infty} \frac{1}{\lambda_j^s} \phi_j \langle \phi_j, f \rangle \right) \left(\sum_{i=1}^{\infty} \phi_i \langle \phi_i, g \rangle \right) d\mu \\
&= \sum_{j=1}^{\infty} \int_K \frac{1}{\lambda_j^s} \phi_j \langle \phi_j, f \rangle \left(\sum_{i=1}^{\infty} \phi_i \langle \phi_i, g \rangle \right) d\mu \\
&= \sum_{j=1}^{\infty} \int_K \frac{1}{\lambda_j^s} \phi_j^2 \langle \phi_j, f \rangle \langle \phi_j, g \rangle d\mu \\
&= \sum_{j=1}^{\infty} \left[\int_K \frac{1}{\lambda_j^s} \phi_j \langle \phi_j, f \rangle \phi_j \langle \phi_j, g \rangle d\mu + \int_K \frac{1}{\lambda_j^s} \phi_j \langle \phi_j, g \rangle \left(\sum_{i=1, i \neq j}^{\infty} \phi_i \langle \phi_i, f \rangle \right) d\mu \right] \\
&= \sum_{j=1}^{\infty} \int_K \left(\frac{1}{\lambda_j^s} \phi_j \langle \phi_j, g \rangle \right) \left(\sum_{i=1}^{\infty} \phi_i \langle \phi_i, f \rangle \right) d\mu \\
&= \int_K \left(\sum_{j=1}^{\infty} \frac{1}{\lambda_j^s} \phi_j \langle \phi_j, g \rangle \right) f \, d\mu \\
&= \langle f, (-\Delta)^s g \rangle
\end{aligned}$$

as desired. $\square$

While the definition of the fractional Laplacian is not the most complicated, we are able to represent $(-\Delta)^{-s}$ in another way, namely with the Riesz kernel.

Definition 2.34: For parameter $s \geq 0$ we define the *fractional Riesz kernel* G_s by

$$G_s(x, y) = \frac{1}{\Gamma(s)} \int_0^\infty t^{s-1} p_t(x, y) \, dt, \quad x, y \in K, x \neq y.$$

Remark 2.35: It follows from (2.8) that $G_s(x, y)$ converges for every $s \geq 0$ with $x \neq y$.

The Riesz kernel allows us to use the estimates of the heat kernel $p_t(x, y)$ in conjunction with $(-\Delta)^{-s}$ as we shall shortly see. We will start by showing some of the bounds of the fractional Riesz kernel.

Proposition 2.36: Let $G_s(x, y)$ be as in Definition 2.34 then the following bounds hold.

(i) If $s \in [0, d_h/d_w)$, there exists a constant $C > 0$ such that for every $x, y \in K, x \neq y$

$$G_s(x, y) \leq \frac{C}{d(x, y)^{d_h - s d_w}}.$$

(ii) If $s = d_h/d_w$, there exists a constant $C > 0$ such that for every $x, y \in K, x \neq y$

$$G_s(x, y) \leq C \left| \ln(d(x, y)) \right|.$$

(iii) If $s > d_h/d_w$, there exists a constant $C > 0$ such that for every $x, y \in K$

$$G_s(x, y) \leq C.$$

Before proving Proposition 2.36 we will quickly state and prove a lemma that is needed.

Lemma 2.37: There exists a constant $C > 0$ such that for every $x, y \in K$ and $t \geq 1$,

$$|p_t(x, y)| \leq C e^{-\lambda_1 t}$$

where $\lambda_1 > 0$ is the first non-zero eigenvalue of $-\Delta$.

Proof. We have from Theorem 2.29 that $p_t(x, y)$ has the expansion

$$p_t(x, y) = \sum_{j=1}^{\infty} e^{-\lambda_j t} \phi_j(x) \phi_j(y). \quad (2.10)$$

Since ϕ_j are eigenvalues, we have from definition of p_t , that for any $t > 0$ we have

$$\phi_j(x) = e^{\lambda_j t} \int_K p_t(x, y) \phi_j(y) \, d\mu(y).$$

Then, from the Cauchy-Schwartz inequality, we have for every $t > 0$

$$|\phi_j(x)| \leq e^{\lambda_j t} \left(\int_K p_t(x, y)^2 \, d\mu(y) \right)^{1/2} \left(\int_K \phi_j(y)^2 \, d\mu(y) \right)^{1/2} = e^{\lambda_j t} p_{2t}(x, x)^{1/2}.$$

The last equality comes from the fact that $\int_K \phi_i(y) \phi_j(y) \, d\mu = \delta_{ij}$ so $\int_K \phi_j(y)^2 \, d\mu(y) = 1$. That combined with (2.10) gives us $\int_K p_t(x, y)^2 \, d\mu(y) = \int_K p_{2t}(x, y) \, d\mu(y)$ resulting in the equality.

Now choosing $t = 1/4$ together with (2.8), we have that there is a constant $C > 0$ so that for every $x \in K$ we have

$$|\phi_j| \leq C e^{\lambda_j/4}.$$

Returning to (2.10) we have for every $x, y \in K$ and $t \geq 1$ that

$$\begin{aligned} |p_t(x, y)| &\leq \sum_{j=1}^{\infty} e^{-\lambda_j t} |\phi_j(x)| |\phi_j(y)| \\ &\leq C^2 \sum_{j=1}^{\infty} e^{-\lambda_j t} e^{\lambda_j/2} \\ &= C^2 \sum_{j=1}^{\infty} e^{-\lambda_j t} e^{\lambda_j/2} e^{\lambda_j} e^{-\lambda_j} \\ &\leq C^2 e^{-\lambda_1 t} e^{\lambda_1} \sum_{j=1}^{\infty} e^{-\lambda_j/2}, \end{aligned}$$

where the last inequality comes from the fact that $0 < \lambda_1 < \lambda_2 < \dots$ and $t \geq 1$ so $e^{\lambda_j(1-t)} \leq e^{\lambda_1(1-t)}$. Thus we have the desired bound. $\square$

Proof of Proposition 2.36. We have that

$$\begin{aligned} G_s(x, y) &= \frac{1}{\Gamma(s)} \int_0^{\infty} t^{s-1} p_t(x, y) dt \\ &= \frac{1}{\Gamma(s)} \int_0^1 t^{s-1} p_t(x, y) dt + \frac{1}{\Gamma(s)} \int_1^{\infty} t^{s-1} p_t(x, y) dt. \end{aligned}$$

The integral $\frac{1}{\Gamma(s)} \int_1^{\infty} t^{s-1} p_t(x, y) dt$ can be uniformly bounded by a constant using Lemma 2.37. Thus we only need to bound the integral $\int_0^1 t^{s-1} p_t(x, y) dt$. By (2.8) we have that

$$\int_0^1 t^{s-1} p_t(x, y) dt \leq c_1 \int_0^1 t^{s-1-d_h/d_w} \exp \left(-c_2 \left(\frac{d(x, y)^{d_w}}{t} \right)^{\frac{1}{d_w-1}} \right) dt.$$

Since the bound of the integral relies on the value of s we will split it up into the cases of Proposition 2.36.

If $s > d_h/d_w$ then we simply have

$$\int_0^1 t^{s-1-d_h/d_w} \exp \left(-c_2 \left(\frac{d(x, y)^{d_w}}{t} \right)^{\frac{1}{d_w-1}} \right) dt \leq \int_0^1 t^{s-1-d_h/d_w} dt.$$

Next, if $s < d_h/d_w$, then using a change of variable $t = u d(x, y)^{d_w}$, we find that

$$\begin{aligned} &\int_0^1 t^{s-1-d_h/d_w} \exp \left(-c_2 \left(\frac{d(x, y)^{d_w}}{t} \right)^{\frac{1}{d_w-1}} \right) dt \\ &= d(x, y)^{s d_w - d_h} \int_0^{1/d(x, y)^{d_w}} u^{s-1-d_h/d_w} \exp \left(-c_2 \left(\frac{1}{u} \right)^{\frac{1}{d_w-1}} \right) du \\ &\leq d(x, y)^{s d_w - d_h} \int_0^{\infty} u^{s-1-d_h/d_w} \exp \left(-c_2 \left(\frac{1}{u} \right)^{\frac{1}{d_w-1}} \right) du. \end{aligned}$$

Convergence of the last integral comes from the fact that $e^{-c_2(1/u)^{1/d_w}}$ converges faster than $1/u$ diverges.

Lastly for $s = d_h/d_w$, we again use the change of variable $t = ud(x, y)^{d_w}$, setting $R = \text{diam}(K)$ and noting that $0 < 1/R^{d_w} < 1/d(x, y)^{d_w}$, we thus find

$$\begin{aligned}
& \int_0^1 t^{s-1-d_h/d_w} \exp \left(-c_2 \left(\frac{d(x, y)^{d_w}}{t} \right)^{\frac{1}{d_w-1}} \right) dt \\
&= \int_0^{1/d(x, y)^{d_w}} u^{-1} \exp \left(-c_2 \left(\frac{1}{u} \right)^{\frac{1}{d_w-1}} \right) du \\
&\leq \int_{1/R^{d_w}}^{1/d(x, y)^{d_w}} u^{-1} \exp \left(-c_2 \left(\frac{1}{u} \right)^{\frac{1}{d_w-1}} \right) du + \int_0^{1/R^{d_w}} u^{-1} \exp \left(-c_2 \left(\frac{1}{u} \right)^{\frac{1}{d_w-1}} \right) du \\
&\leq \int_{1/R^{d_w}}^{\frac{1}{d(x, y)^{d_w}}} u^{-1} du + \int_0^{1/R^{d_w}} u^{-1} \exp \left(-c_2 \left(\frac{1}{u} \right)^{\frac{1}{d_w-1}} \right) du \\
&\leq d_w |\ln(d(x, y))| + \int_0^{1/R^{d_w}} u^{-1} \exp \left(-c_2 \left(\frac{1}{u} \right)^{\frac{1}{d_w-1}} \right) du \\
&\leq C |\ln(d(x, y))|.
\end{aligned}$$

Thus we have our desired bounds. $\square$

2.7 Bounds of the Fractional Laplacian

In the following we will prove some bounds for the fractional Laplacian. Since the goal of this thesis is to prove convergence of fractional Gaussian fields within a specific space of test functions, we will only be proving the lemmas for that space. The space of functions is given by

Definition 2.38: We define the following space of test functions $S(K)$ as

$$S(K) = \left\{ f \in C_0(K) : \forall k \geq 0 \lim_{n \rightarrow \infty} n^k \left| \int_K \phi_n(y) f(y) d\mu(y) \right| = 0 \right\}$$

with the topology defined by the family of norms

$$\|f\|_k = \|(-\Delta)^{-k} f\|_{L^2(K, \mu)}.$$

By [1, Theorem 2] and Lemma 2.30 we have that $S(K)$ is a Fréchet nuclear space. That a space is *Fréchet* means that it is nuclear and the topology is given by a countable family of seminorms. We will quickly define a nuclear space, but will not delve further into the subject matter. The following definition is from [2, Chapter 50].

Definition 2.39: Let E be a locally convex Hausdorff space and $\|\cdot\|_p$ a continuous seminorm on E . Furthermore let $\widehat{E}_p$ be a Banach space, defined to be the completion of the normed space $E \setminus \ker \|\cdot\|_p$. Then E is a *nuclear space* if for every continuous seminorm $\|\cdot\|_p$ on E there exists another continuous seminorm on E with $\|\cdot\|_q \geq \|\cdot\|_p$ such that the canonical mapping $\widehat{E}_p \rightarrow \widehat{E}_q$ is *nuclear*.

A mapping $u: E \rightarrow F$, where E, F are Banach spaces, is *nuclear* if there is a sequence $\{x'_k\}$ in the closed unit ball of E' , a sequence $\{y_k\}$ in the closed unit ball of F , and a complex sequence $\{\lambda_k\}$ with $\sum_k |\lambda_k| < \infty$ such that $u(x) = \sum_{i=1}^{\infty} \lambda_i x'_i(x) y_i$ for all $x \in E$.

Furthermore, the trace-norm $\|u\|_{\text{Tr}}$ is equal to the infimum of the numbers $\sum_k |\lambda_k|$ over the set of all representations of u as such a series.

Remark 2.40: We have just defined a nuclear map on a Banach spaces, but note that the definition can be generalized to locally convex Hausdorff spaces.

The concept of a nuclear space is quite complicated, but its properties are very useful. It will allow us to prove the convergence of fractional Gaussian fields through the characteristics function and existence of fractional Gaussian fields through Bochner-Minlos theorem for nuclear spaces, which will be state later in Theorem 3.5. But first we will prove some lemmas regarding the fractional Laplacian and the fractional Riesz kernel.

Lemma 2.41: Let $s > 0$. For $f \in S(K)$ and $x \in K$

$$(-\Delta)^{-s} f = \int_K G_s(x, y) f(y) \, d\mu(y).$$

Proof. First, we will make sure the integral on the right side actually converges. This holds because $S(K) \subset C(K)$ hence f will be bounded on K . Thus we just need for every $x \in K$, that $\int_K G_s(x, y) \, d\mu(y)$ is convergent.¹ For $s \in (d_h/d_w, \infty)$ the convergence follows from proposition 2.36. Looking at $s \in (0, d_h/d_w]$, we will use a dyadic annuli decomposition to split up K into infinitely many annuli on the form $B(x, R2^{-j}) \setminus B(x, RS^{-j-1})$, noting that

$$K \subseteq \bigcup_{j \geq 0} B(x, R2^{-j}) \setminus B(x, RS^{-j-1}),$$

where $R = \text{diam}(K)$. For all $y \in B(x, R2^{-j}) \setminus B(x, RS^{-j-1})$ we then have that

$$R2^{-j-1} \leq d(x, y) \leq R2^{-j-1} \Rightarrow 2^{-j-1} \leq R^{-1} \cdot d(x, y) \leq 2^{-j-1}.$$

We will show that the integral is finite for both $s = d_h/d_w$ and $s < d_h/d_w$, but this requires that we bound $|\ln(d(x, y))|$. So we see that for all $\alpha > 0$

$$x^\alpha |\ln(x)| \rightarrow 0, \quad \text{as } x \rightarrow 0,$$

which means there exists a constant $k > 0$ such we can choose some interval $(0, u_k]$ so $\ln(x) \leq \frac{k}{x^\alpha}$ for all $x \in (0, u_k]$.

¹In Proposition 2.36, (i) and (ii) the bounds are for $x \neq y$, but they are null-sets with respect to μ

Now choose k' such $u_{k'} \leq R$, in which case for $s = d_h/d_w$ we have

$$\begin{aligned} \int_K G_s(x, y) \, d\mu(y) &\leq \int_K C_1 \cdot |\ln(d(x, y))| \, d\mu(y) \\ &\leq \sum_{j=0}^{\infty} \int_{B(x, R2^{-j}) \setminus B(x, RS^{-j-1})} C_1 \cdot |\ln(d(x, y))| \, d\mu(y) \\ &\leq \sum_{j=0}^{\infty} \int_{B(x, R2^{-j}) \setminus B(x, RS^{-j-1})} \frac{C_1 \cdot k'}{d(x, y)^\gamma} \, d\mu, \end{aligned}$$

where $\gamma < d_h$. We can likewise bound $G_s(x, y)$ by

$$\begin{aligned} \int_K G_s(x, y) \, d\mu(y) &\leq \int_K \frac{C_2}{d(x, y)^\gamma} \, d\mu(y) \\ &\leq \sum_{j=0}^{\infty} \int_{B(x, R2^{-j}) \setminus B(x, RS^{-j-1})} \frac{C}{d(x, y)^\gamma} \, d\mu(y), \end{aligned}$$

where $C = \max\{C_1 \cdot k', C_2\}$. Thus for $s \in (0, d_h/d_w]$ we have

$$\begin{aligned} \int_K G_s(x, y) \, d\mu(y) &\leq \sum_{j=0}^{\infty} \int_{B(x, R2^{-j}) \setminus B(x, RS^{-j-1})} \frac{C}{d(x, y)^\gamma} \, d\mu(y) \\ &\leq R \cdot C \sum_{j=0}^{\infty} 2^{j\gamma} \mu(B(x, R2^{-j}) \setminus B(x, RS^{-j-1})) \\ &\leq R \cdot C \sum_{j=0}^{\infty} 2^{j\gamma} \mu(B(x, R2^{-j})) \\ &\leq R^{d_h+1} \cdot C \cdot c_2 \sum_{j=0}^{\infty} 2^{j(\gamma-d_h)} < \infty, \end{aligned}$$

where we use proposition 1.16. Therefore $\int_K G_s(x, y) \, d\mu(y)$ is convergent for every $x \in K$.

Moving along in the proof, we note that if we use the substitution $t = \lambda_j u$ we get

$$\frac{1}{\Gamma(s)} \int_0^\infty t^{s-1} e^{-\lambda_j t} \, dt = \frac{1}{\Gamma(s) \cdot \lambda_j^s} \int_0^\infty u^{s-1} e^{-u} \, du = \frac{1}{\Gamma(s) \lambda_j^s} \Gamma(s) = \lambda_j^{-s}.$$

Now using Fubini's theorem, Definition 2.34 and Theorem 2.29 we get

$$\begin{aligned} \int_K G_s(x, y) f(y) \, d\mu(y) &= \int_K \frac{1}{\Gamma(s)} \int_0^\infty t^{s-1} p_t(x, y) \, dt f(y) \, d\mu(y) \\ &= \int_K \frac{1}{\Gamma(s)} \int_0^\infty t^{s-1} \sum_{j=1}^{\infty} e^{-\lambda_j t} \phi_j(x) \phi_j(y) \, dt f(y) \, d\mu(y) \\ &= \sum_{j=1}^{\infty} \phi_j(x) \int_K \phi_j(y) f(y) \frac{1}{\Gamma(s)} \int_0^\infty t^{s-1} e^{-\lambda_j t} \, dt \, d\mu(y) \\ &= \sum_{j=1}^{\infty} \frac{1}{\lambda_j^s} \phi_j(x) \int_K \phi_j(y) f(y) \, d\mu(y) \\ &= (-\Delta)^{-s} f \end{aligned}$$

as desired. □

An immediate corollary follows:

Corollary 2.42: Let $(-\Delta)^{-s}$ be defined as in Definition 2.31, then the following holds

(i) Let $s > 0$, for $f, g \in S(K)$

$$\int_K (-\Delta)^{-s} f (-\Delta)^{-s} g \, d\mu = \int_K \int_K f(x) g(y) G_{2s}(x, y) \, d\mu(x) \, d\mu(y).$$

(ii) Let $s > 0$ the for $f \in S(K)$

$$(-\Delta)^{-s} f = \frac{1}{\Gamma(s)} \int_0^\infty t^{s-1} P_t f(x) \, dt.$$

Proof. Beginning with (i). It follows from Proposition 2.33 and Lemma 2.41 that

$$\begin{aligned} \int_K (-\Delta)^{-s} f (-\Delta)^{-s} g \, d\mu &= \langle (-\Delta)^{-s} f, (-\Delta)^{-s} g \rangle \\ &= \langle f, (-\Delta)^{-2s} g \rangle \\ &= \int_K f (-\Delta)^{-2s} g \, d\mu \\ &= \int_K \int_K f(x) g(y) G_{2s}(x, y) \, d\mu(x) \, d\mu(y). \end{aligned}$$

Moving on to (ii). Again from Lemma 2.41 and Theorem 2.29 we have

$$\begin{aligned} (-\Delta)^{-s} f &= \sum_{j=1}^{\infty} (\lambda_j)^{-s} \phi_j \int_K \phi_j(y) f(y) \, d\mu(y) \\ &= \sum_{j=1}^{\infty} \left(\frac{1}{\Gamma(s)} \int_0^\infty t^{s-1} e^{-\lambda_j t} \, dt \right) \phi_j \int_K \phi_j(y) f(y) \, d\mu(y) \\ &= \frac{1}{\Gamma(s)} \int_0^\infty t^{s-1} \int_K \sum_{j=1}^{\infty} e^{-\lambda_j t} \phi_j \phi_j(y) f(y) \, d\mu(y) \, dt \\ &= \frac{1}{\Gamma(s)} \int_0^\infty t^{s-1} \int_K p_t(x, y) f(y) \, d\mu(y) \, dt \\ &= \frac{1}{\Gamma(s)} \int_0^\infty t^{s-1} P_t f(x) \, dt. \end{aligned}$$

□

Chapter 3

Fractional Gaussian Fields

In this chapter we will be constructing both the fractional Gaussian fields and the discrete fractional Gaussian fields, that we will later see converge in distribution. The construction of the fractional Gaussian fields very much depends on the Riesz kernel and the Dirichlet Laplacian, with the discrete fractional Gaussian field depending on the discrete fractional Laplacian and discrete Riesz kernel.

We will also give an example of a discrete fractional Gaussian field, which we will use for simulations in Chapter 5.

3.1 Discrete Fractional Gaussian Fields

Definition 3.1: Let $s \geq 0$. A discrete fractional Gaussian field X_s^m with parameter s on V_m is a Gaussian vector indexed by V_m with mean 0 and covariance matrix $\Sigma = (G_{2s}^m(x, y))_{x, y \in V_m}$.

Notation: For any $f \in \ell(V_m)$ we will use the notation

$$X_s^m(f) = \frac{1}{a_m} \sum_{p \in V_m} f(p) X_s^m(p).$$

We then note that discrete fractional Gaussian fields has the following property:

Proposition 3.2: For $f, g \in \ell(V_m)$ we have that

$$\mathbb{E} [X_s^m(f) X_s^m(g)] = \frac{1}{a_m} \sum_{p \in V_m} (-\Delta_m)^{-s} f(p) (-\Delta_m)^{-s} g(p).$$

Proof. Using Proposition 2.23 we find

$$\begin{aligned} \mathbb{E} [X_s^m(f) X_s^m(g)] &= \frac{1}{2a_m} \sum_{p, q \in V_m} f(p) g(q) \mathbb{E} [X_s^m(p) X_s^m(q)] \\ &= \frac{1}{2a_m} \sum_{p, q \in V_m} f(p) g(q) G_{2s}^m(p, q) \\ &= \frac{1}{a_m} \sum_{p \in V_m} f(p) \left(\frac{1}{a_m} \sum_{q \in V_m} g(q) G_{2s}^m(p, q) \right) \\ &= \frac{1}{a_m} \sum_{p \in V_m} f(p) (-\Delta_m)^{-2s} g(p) \\ &= \frac{1}{a_m} \sum_{p \in V_m} (-\Delta_m)^{-s} f(p) (-\Delta_m)^{-s} g(p). \end{aligned}$$

□

Example 3.3: Here we will give an example of a discrete fractional Gaussian field. Let $(W_i)_{1 \leq i \leq N_m}$ be a sequence of i.i.d. Gaussian random variables with mean 0 and variance 1, i.e. white noise. Then define

$$X_s^m(x) = \sum_{i=1}^{N_m} (\lambda_i^m)^{-s} \phi_i^m(x) W_i, \quad x \in V_m.$$

Initially, we check to see that the mean

$$\mathbb{E}[X_s^m] = \begin{pmatrix} \mathbb{E}[X_s^m(x_1)] \\ \vdots \\ \mathbb{E}[X_s^m(x_{N_m})] \end{pmatrix} = \begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix},$$

since

$$\mathbb{E}[X_s^m(x_j)] = \mathbb{E}\left[\sum_{i=1}^{N_m} (\lambda_i^m)^{-s} \phi_i^m(x_j) W_i\right] = \sum_{i=1}^{N_m} (\lambda_i^m)^{-s} \phi_i^m(x_j) \mathbb{E}[W_i] = 0.$$

Moving on to the covariance matrix, we have that

$$\begin{aligned} \Sigma &= \mathbb{E}\left((X_s^m - \mathbb{E}[X_s^m])((X_s^m)^T - \mathbb{E}[X_s^m]^T)\right) \\ &= \mathbb{E}\left[X_s^m \cdot (X_s^m)^T\right] = \mathbb{E}\left[\begin{pmatrix} X_s^m(x_1)X_s^m(x_1) & \cdots & X_s^m(x_1)X_s^m(x_{N_m}) \\ \vdots & \ddots & \vdots \\ X_s^m(x_{N_m})X_s^m(x_1) & \cdots & X_s^m(x_{N_m})X_s^m(x_{N_m}) \end{pmatrix}\right]. \end{aligned}$$

Looking at a single element $\Sigma_{p,q}$, and using that W_i are independent, we compute

$$\begin{aligned} \mathbb{E}[X_s^m(x_p)X_s^m(x_q)] &= \mathbb{E}\left[\left(\sum_{i=1}^{N_m} (\lambda_i^m)^{-s} \phi_i^m(x_p) W_i\right) \left(\sum_{j=1}^{N_m} (\lambda_j^m)^{-s} \phi_j^m(x_q) W_j\right)\right] \\ &= \mathbb{E}\left[\sum_{i=1}^{N_m} (\lambda_i^m)^{-2s} \phi_i^m(x_p) \phi_i^m(x_q) W_i^2\right] \\ &\quad + \mathbb{E}\left[\sum_{i,j=1, i \neq j}^{N_m} (\lambda_i^m \lambda_j^m)^{-s} \phi_i^m(x_p) \phi_j^m(x_q) W_i W_j\right] \\ &= \sum_{i=1}^{N_m} (\lambda_i^m)^{-2s} \phi_i^m(x_p) \phi_i^m(x_q) \mathbb{E}[W_i^2] \\ &\quad + \sum_{i,j=1, i \neq j}^{N_m} (\lambda_i^m \lambda_j^m)^{-s} \phi_i^m(x_p) \phi_j^m(x_q) \mathbb{E}[W_i] \mathbb{E}[W_j] \\ &= \sum_{i=1}^{N_m} (\lambda_i^m)^{-2s} \phi_i^m(x_p) \phi_i^m(x_q) \\ &= G_{2s}^m(x_p, x_q). \end{aligned}$$

Thus X_s^m is a discrete fractional Gaussian field. ◇

3.2 Fractional Gaussian Fields

Definition 3.4: Let $s \geq 0$. A *fractional Gaussian field* X_s with parameter s on K is a centered Gaussian field, $\{X_s(f) : f \in S(K)\}$, such that for any $f, g \in S(K)$

$$\mathbb{E}[X_s(f)X_s(g)] = \int_K (-\Delta)^{-s} f (-\Delta)^{-s} g \, d\mu.$$

We now move on to proving that the fractional Gaussian fields exist. First we will state the Bochner-Minlos theorem for nuclear spaces, since it is needed for the proof of existence. It will be given without proof but one can be found in [14, Theorem 2.3.1].

Theorem 3.5 (Bochner-Minlos): Assume $\mathcal{X}$ is a nuclear space. Any continuous positive-definite linear functional Λ on $\mathcal{X}$ satisfying $\Lambda(0) = 1$ is the Fourier transform of a countably additive positive normalized measure on $\mathcal{X}'$.

In the next lemma we will prove that a functional, which we will need, is positive semi-definite.

Lemma 3.6: The functional $\Phi: \mathcal{X} \rightarrow \mathbb{R}$ defined as $\Phi(v) = \exp\left(-\frac{1}{2} \langle v, v \rangle\right)$ is positive definite.

Proof. Let $v_1, \dots, v_n$ be elements of $\mathcal{X}$ and further choose an orthonormal basis $e_1, \dots, e_m$ of $\text{span}\{v_1, \dots, v_n\}$. Additionally let $Z = (Z_1, \dots, Z_m)$ be a vector of independent standard normal random variables, and note that for all $u \in \mathbb{R}^m$ we have that

$$\Phi\left(\sum_{j=1}^m u_j e_j\right) = \exp\left(-\frac{1}{2} \sum_{i=1}^m u_i^2\right) = \mathbb{E}\left[e^{iu \cdot Z}\right].$$

This leads us to for $z \in \mathbb{C}^n$

$$\begin{aligned} \sum_{j,k=1}^n z_j \overline{z_k} \Phi(v_j - v_k) &= \sum_{j,k=1}^n z_j \overline{z_k} \mathbb{E}\left[e^{i(v_j - v_k) \cdot Z}\right] \\ &= \mathbb{E}\left[\left(\sum_{j=1}^n z_j e^{iv_j \cdot Z}\right) \left(\sum_{k=1}^n \overline{z_k} e^{iv_k \cdot Z}\right)\right] = \mathbb{E}\left[\left|\sum_{j=1}^n z_j e^{iv_j \cdot Z}\right|^2\right] \geq 0 \end{aligned}$$

and ϕ is positive semidefinite. □

We can now give a proof that fractional Gaussian fields exist.

Theorem 3.7: Let $s \geq 0$. There exists a centered Gaussian distribution X_s on $S'(K)$ such that for $f, g \in S(K)$

$$\mathbb{E}[X_s(f)X_s(g)] = \int_K (-\Delta)^{-s} f (-\Delta)^{-s} g \, d\mu.$$

Proof. The space $S(K)$ is a nuclear space, why it is enough to use Theorem 3.5 (Bochner-Minlos) on the functional

$$\Lambda: f \rightarrow \exp\left(-\frac{1}{2}\|f\|_k^2\right).$$

Since if Theorem 3.5 holds, then it gives us a unique probability measure ν such that

$$\exp\left(-\frac{1}{2}\|f\|_k^2\right) = \int_{S'(K)} e^{i\langle x, f \rangle} d\nu(x).$$

Notice that the right side is a characteristic function for some random variable. If we choose $f = t \cdot g$ for some $g \in S(K)$ and $t \in \mathbb{R}$, noting that $t \cdot g \in S(K)$, then the left side of the equation becomes the characteristic function for the normal distribution with mean 0 and variance $\|f\|_s^2$. Thus we have the X_s we wanted.

Now we prove that Λ satisfies the requirements of Theorem 3.5. Obviously $\Lambda(0) = 1$, and from lemma 3.6 Λ is positive definite. Lastly we need to prove that it is continuous at 0.

First we note that since $(-\Delta)^{-s}$ is a bounded operator we have that there exists some constant $C > 0$ so $\|(-\Delta)^{-s}f\|_{L^2(K, \mu)} \leq C\|f\|_{L^2(K, \mu)}$. Now let $\varepsilon > 0$ be given and choose

$$\delta = \begin{cases} \left(\frac{-2 \cdot \ln(1-\varepsilon)}{C}\right)^{1/2}, & \text{when } \varepsilon \in (0, 1) \\ 1, & \text{when } \varepsilon \geq 1. \end{cases}$$

If $\varepsilon \geq 1$ notice that

$$\left| \exp\left(-\frac{1}{2}\|f\|_s^2\right) - \exp\left(-\frac{1}{2}\|0\|_s^2\right) \right| \leq 1 \leq \varepsilon, \quad \forall f \in S(K)$$

in which case we can choose δ to be any real number larger than 0. Moving on to $\varepsilon \in (0, 1)$ we get

$$\begin{aligned} \left| \exp\left(-\frac{1}{2}\|f\|_s^2\right) - \exp\left(-\frac{1}{2}\|0\|_s^2\right) \right| &\leq \left| \exp\left(-\frac{1}{2}C\|f\|_{L^2(K, \mu)}^2\right) - 1 \right| \\ &= \left| \exp\left(-\frac{1}{2}C \int_K |f|^2 d\mu\right) - 1 \right| \\ &\leq \left| \exp\left(-\frac{1}{2}C \int_K \delta^2 d\mu\right) - 1 \right| \\ &= \left| \exp\left(-\frac{1}{2}C \int_K \frac{-2 \cdot \ln(1-\varepsilon)}{C} d\mu\right) - 1 \right| \\ &= \left| \exp(\ln(1-\varepsilon)) - 1 \right| \\ &= \varepsilon, \end{aligned}$$

using that μ is a probability measure in the second to last equality. Thus we have continuity in 0 as desired. $\square$

In the following we want to prove that a fractional Gaussian field has a L^2 density if and only if $s > \frac{d_h}{2d_w}$, but for this we need white noise.

Definition 3.8: Let $(X, \mathcal{E}, \mu)$ be σ -finite measure space. A *white noise based on ν* is a random set function W on the sets $A \in \mathcal{E}$ with $\nu(A) < \infty$ such that

- (i) $W(A)$ is a $N(0, \mu(A))$ random variable.
- (ii) If $A \cap B = \emptyset$ then $W(A)$ and $W(B)$ are independent and $W(A \cup B) = W(A) + W(B)$.

We will now show that such a process exists on $(K, \mathcal{K}, \mu)$ where $\mathcal{K}$ is the Borel σ -algebra on K .

We can think of white noise as a Gaussian process indexed by the set

$$\mathcal{A} = \{W(A) : A \in \mathcal{K}, \mu(A) < \infty\}.$$

From Definition 3.8, (i) and (ii) it is a mean-zero Gaussian process with covariance matrix

$$\Sigma = \left(\mathbb{E} [W(A)W(B)] \right)_{W(A), W(B) \in \mathcal{A}} = (\mu(A \cap B))_{W(A), W(B) \in \mathcal{A}}.$$

Now let $A_1, \dots, A_n \in \mathcal{K}$ and $a_1, \dots, a_n$ be real numbers. Then

$$\sum_{i,j=1}^n a_i a_j \Sigma_{i,j} = \sum_{i,j=1}^n a_i a_j \int_K \mathbb{1}_{A_i} \cdot \mathbb{1}_{A_j} d\mu = \int_K \left(\sum_{i=1}^n a_i \mathbb{1}_{A_i} \right)^2 d\mu \geq 0$$

Thus Σ is positive definite, so there exists a probability space $(\Omega, \mathcal{F}, p)$ and a mean-zero Gaussian process $\{W(A)\}$ on $(\Omega, \mathcal{F}, p)$ such that W satisfies Definition 3.8 (i) and (ii).

This means that for $f \in L^2(K, \mathcal{K}, \mu)$ the integral with respect to W , $W(f) = \int_K f dW$ is a well defined Gaussian random variable. Moreover, the Gaussian measure W gives rise to an isonormal Gaussian family, $\{W(f), f \in L^2(K, \mathcal{K}, \mu)\}$, with the covariance function

$$\mathbb{E} [W(f)W(g)] = \int_K fg d\mu.$$

In the following we want to define a fractional Brownian field, and we want to do it as the integral of the fractional Riesz kernel, which was defined in Definition 2.34, thus we need $G_s(x, \cdot) \in L^2(K, \mu)$.

Let $s > \frac{d_h}{2d_w}$. We observe from Proposition 2.36 that $G_s(x, \cdot) \in L^2(K, \mu)$. Moreover we have that

$$\begin{aligned} G_s(x, y) &= \frac{1}{\Gamma(s)} \int_K t^{s-1} p_t(x, y) dt \\ &= \frac{1}{\Gamma(s)} \int_K t^{s-1} \sum_{j=1}^{\infty} e^{-\lambda_j t} \phi_j(x) \phi_j(y) dt = \sum_{j=1}^{\infty} \frac{1}{\lambda_j^{-s}} \phi_j(x) \phi_j(y). \end{aligned} \quad (3.1)$$

recalling Equation (2.9) and $\frac{1}{\Gamma(s)} \int_K t^{s-1} e^{-\lambda_j t} dt = \lambda_j^{-s}$. Furthermore note that by Lemma 2.30 we have (3.1) converges, since $s > \frac{d_h}{2d_w}$.

Definition 3.9: Let $s > \frac{d_h}{2d_w}$. The *fractional Brownian field* with parameter s is defined as

$$\tilde{X}_s(x) = \int_K G_s(x, y) \, dW(y).$$

Remark 3.10: Since $s > \frac{d_h}{2d_w}$ then by (3.1) we can write

$$\tilde{X}_s(x) = \sum_{i=1}^{\infty} \lambda_i^{-s} \phi_i(x) W_i,$$

where $W_i = W(\phi_i)$ is an i.i.d. sequence of Gaussian random variables with mean 0 and variance 1.

We will now show that one can construct a fractional Gaussian field from the fractional Brownian field.

Proposition 3.11: Let $s > \frac{d_h}{2d_w}$. Then the Gaussian random field defined by

$$X_s(f) = \int_K f(x) \tilde{X}_s(x) \, d\mu(x), \quad f \in S(K)$$

has the law of a fractional Gaussian field with parameter s . Moreover, if there exists a Gaussian field $(\tilde{Y}_s(x))_{x \in K}$ on K with

$$\mathbb{E} \left[\int_K \tilde{Y}_s(x)^2 \, d\mu(x) \right] < \infty$$

such that the Gaussian field defined by

$$Y_s(f) = \int_K f(x) \tilde{Y}_s(x) \, d\mu(x), \quad f \in S(K)$$

has the law of fractional Gaussian field with parameter s , then $s > \frac{d_h}{2d_w}$.

Proof. Assume $s > \frac{d_h}{2d_w}$. Then from Fubini's theorem we have for every $f \in S(K)$ a.s. that

$$\int_K f(x) \tilde{X}_s(x) \, d\mu(x) = \int_K f(x) \int_K G_s(x, y) \, dW(y) \, d\mu(x) = \int_K (-\Delta)^{-s} f(y) \, dW(y),$$

in which case by definition of W we have that $X_s(f) = \int_K f(x) \tilde{X}_s(x) \, d\mu(x)$ is a Gaussian random variable with mean 0 and variance

$$\int_K |(-\Delta)^{-s} f(x)|^2 \, d\mu(x)$$

and is thus a fractional Gaussian field with parameter s .

Now assume that there is a Gaussian field $(\tilde{Y}_s(x))_{x \in K}$ on K with

$$\mathbb{E} \left[\int_K \tilde{Y}_s(x)^2 \, d\mu(x) \right] < \infty,$$

such that the Gaussian field defined by

$$Y_s(f) = \int_K f(x) \tilde{Y}_s(x) \, d\mu(x), \quad f \in S(K), \quad (3.2)$$

has the law of fractional Gaussian field with parameter s . Then by spectral decomposition and (3.2) we have

$$\tilde{Y}_s(x) = \sum_{i=1}^{\infty} \langle \tilde{Y}_s, \phi_i \rangle \phi_i(x) = \sum_{i=1}^{\infty} Y_s(\phi_i) \phi_i(x).$$

Note that $(Y(\phi_i))_{i \geq 1}$ is a sequence of independent Gaussian random variables with mean 0 and variance $(\lambda_i^{-2s})_{i \geq 1}$, since by Definition 3.4 and Proposition 2.33 we have that

$$\mathbb{E}[Y_s(\phi_i)] = \mathbb{E} \left[\int_K \phi_i \tilde{Y}_s \, d\mu \right] = \mathbb{E} \left[\int_K \phi_i(x) \sum_{j=1}^{\infty} \lambda_j^{-s} \phi_j(x) W_j \, d\mu(x) \right] = \mathbb{E}[W_j] = 0$$

and

$$\mathbb{E}[Y_s(\phi_i) Y_s(\phi_j)] = \int_K (-\Delta)^{-s} \phi_i (-\Delta)^{-s} \phi_j \, d\mu = \frac{1}{(\lambda_j \lambda_i)^s} \int_K \phi_i \phi_j \, d\mu = \frac{1}{(\lambda_j \lambda_i)^s} \delta_{ij}.$$

Since $\mathbb{E} \left[\int_K \tilde{Y}_s(x)^2 \, d\mu(x) \right] < \infty$ we find that

$$\begin{aligned} \mathbb{E} \left[\int_K \tilde{Y}_s(x)^2 \, d\mu(x) \right] &= \mathbb{E} \left[\int_K \left(\sum_{i=1}^{\infty} Y_s(\phi_i) \phi_i(x) \right)^2 \, d\mu(x) \right] \\ &= \mathbb{E} \left[\sum_{i=1}^{\infty} Y_s(\phi_i)^2 \int_K \phi_i(x) \phi_i(x) \, d\mu(x) \right] \\ &= \sum_{i=1}^{\infty} \mathbb{E} [Y_s(\phi_i)^2] \\ &= \sum_{i=1}^{\infty} \frac{1}{\lambda_i^{2s}} < \infty, \end{aligned}$$

where we use that $\int_K (\sum_{i=1}^{\infty} Y_s(\phi_i) \phi_i(x))^2 \, d\mu(x) = \int_K \sum_{i=1}^{\infty} Y_s(\phi_i)^2 \phi_i(x)^2 \, d\mu(x)$ which stems from the fact that $\int_K \phi_i \phi_j \, d\mu = \delta_{ij}$ and $Y_s(\phi_i)$ is a constant with respect to $\mu(x)$.

We now have that

$$\mathbb{E} \left[\int_K \tilde{Y}_s(x)^2 \, d\mu(x) \right] = \sum_{i=1}^{\infty} \frac{1}{\lambda_i^{2s}} < \infty.$$

in which case by Lemma 2.30 we have that $s > \frac{d_h}{2d_w}$. □

Chapter 4

Convergence in distribution of Fractional Gaussian Fields

Now that we have presented and proved the existence of fractional Gaussian fields X_s on K and discrete fractional Gaussian fields X_s^m on V_m , our next task is proving that they converge. Therefore, in this section our main goal is to show that X_s^m converges in distribution to X_s .

4.1 Preliminary Lemmas

In this section we will prove a collection of lemmas that will be needed.

Lemma 4.1: Let $\{P_t\}_{t \geq 0}$ be the semigroup generated by the Dirichlet Laplacian $(-\Delta)^{-s}$ and $p_t(x, y)$ the heat kernel admitted by $\{P_t\}_{t \geq 0}$. If ϕ_i is an orthonormal eigenfunction of $(-\Delta)^{-s}$ then

$$P_t \phi_i(x) = e^{-\lambda_i t} \phi_i(x).$$

Proof. The lemma follows immediately from these calculations:

$$\begin{aligned} P_t \phi_i(x) &= \int_K p_t(x, y) \phi_i(y) \, d\mu(y) = \int_K \left(\sum_{j=1}^{\infty} e^{-\lambda_j t} \phi_j(x) \phi_j(y) \right) \phi_i(y) \, d\mu(y) \\ &= \int_K e^{-\lambda_i t} \phi_i(x) \, d\mu(y) = e^{-\lambda_i t} \phi_i(x). \end{aligned}$$

□

Lemma 4.2: For all $f \in S(K)$ and $t > 0$,

$$\lim_{m \rightarrow \infty} \frac{1}{a_m} \sum_{p \in V_m} f_m(p) P_t^m f_m(p) = \int_K f(x) P_t f(x) \, d\mu(x)$$

where $f_m = f|_{V_m}$.

Proof. In this proof we wish to use [3, Theorem 2.1] so we will check that Δ fulfils the criteria of [3, Theorem 2.1]. First there is a bounded linear operator such that for all $f \in L^2(K, \mu)$ we have

$$\lim_{n \rightarrow \infty} \|P_n f\| = \|P_n\|.$$

This holds in our case, since we have $P_n f = f|_{V_n} = f_n$.

Secondly we note that Δ is the infinitesimal generator of a uniformly continuous semigroup by Theorem 2.25.

Next we have that the sequence of $\{\Delta_m\}_{m \geq 0}$ indeed coincides with the extended limit defined, in [3, p. 355], this follows from Theorem 2.16 and Section 2.3.

Finally the range $\mathcal{R}(\lambda_0 - \Delta)$ is dense in L^2 by [33, Lemma 1.1.1]. Thus by [3, Theorem 2.1] we have that for all $t \geq 0$

$$\lim_{m \rightarrow \infty} \sup_{p \in V_m} |P_t^m f_m(p) - (P_t f(p))_m| = 0. \quad (4.1)$$

If we now write

$$\frac{1}{a_m} \sum_{p \in V_m} f_m(p) P_t^m f_m(p) = \int_{V_m} f_m P_t^m f_m \, d\mu_m,$$

then

$$\int_{V_m} f_m P_t^m f_m \, d\mu_m = \int_{V_m} f_m (P_t^m f_m - (P_t f)_m) \, d\mu_m + \int_{V_m} f_m (P_t f)_m \, d\mu_m.$$

The first integral converges to 0 as $m \rightarrow \infty$ because of (4.1). Furthermore, by Corollary 1.18 we have that

$$\int_{V_m} f_m (P_t f)_m \, d\mu_m \rightarrow \int_K f P_t f \, d\mu, \quad \text{as } m \rightarrow \infty.$$

□

Lemma 4.3: For all $f \in S(K)$ and $s \geq 0$ the following holds,

$$\frac{1}{a_m} \sum_{p \in V_m} f_m(p) (-\Delta_m)^{-2s} f_m(p) \rightarrow \int_K f(x) (-\Delta)^{-2s} f(x) \, d\mu(x), \quad \text{as } m \rightarrow \infty.$$

Proof. For $s = 0$ we find that

$$\begin{aligned} (-\Delta_m)^{-0} f &= \sum_{i=1}^{N_m} \frac{1}{(\lambda_i^m)^0} \phi_i^m(x) \frac{1}{a_m} \sum_{y \in V_m} \phi_i^m(y) f(y) \\ &= \sum_{i=1}^{N_m} \phi_i^m(x) \int_{V_m} \phi_i^m(y) f(y) \, d\mu_m \\ &= f_m. \end{aligned}$$

Likewise by Definition 2.31 we find

$$(-\Delta)^{-0} f = \sum_{i=1}^{\infty} \phi_i \int_K \phi_i(y) f(y) \, d\mu = f,$$

and the lemma for $s = 0$ follows from Corollary 1.18.

Now assume $s > 0$, then by Remark 2.28 we have that

$$\begin{aligned}
(-\Delta_m)^s f_m(x) &= \frac{1}{a_m} \sum_{j=1}^{N_m} (\lambda_j^m)^{-s} \phi_j^m(x) \sum_{y \in V_m} \phi_j(y) f_m(y) \\
&= \frac{1}{a_m} \sum_{j=1}^{N_m} \frac{1}{\Gamma(s)} \int_0^\infty t^{s-1} e^{-\lambda_j t} dt \phi_j^m(x) \sum_{y \in V_m} \phi_j(y) f_m(y) \\
&= \frac{1}{\Gamma(s)} \int_0^\infty t^{s-1} \sum_{j=1}^{N_m} e^{-\lambda_j t} \phi_j^m(x) \frac{1}{a_m} \sum_{y \in V_m} \phi_j^m(y) f_m(y) dt \\
&= \frac{1}{\Gamma(s)} \int_0^\infty t^{s-1} P_t^m f(x) dt.
\end{aligned}$$

Now by Fubini's theorem we have that

$$\frac{1}{a_m} \sum_{p \in V_m} f_m(p) (-\Delta_m)^{-2s} f_m(p) = \frac{1}{\Gamma(2s)} \int_0^\infty t^{2s-1} \frac{1}{a_m} \sum_{p \in V_m} f_m(p) P_t^m f_m(p) dt$$

By Lemma 4.1 and Corollary 2.26 we find that

$$\begin{aligned}
\frac{1}{a_m} \sum_{p \in V_m} f_m(p) P_t^m f_m(p) &= \frac{1}{a_m} \sum_{p \in V_m} f_m(p) P_t^m \sum_{j=1}^\infty \langle f_m, \phi_j \rangle \phi_j \\
&= \frac{1}{a_m} \sum_{p \in V_m} f_m(p) \sum_{j=1}^\infty \langle f_m, \phi_j \rangle P_t^m \phi_j \\
&= \frac{1}{a_m} \sum_{p \in V_m} f_m(p) \sum_{j=1}^\infty \langle f_m, \phi_j \rangle e^{-\lambda_j^m t} \phi_j \\
&\leq e^{-\lambda_1^m t} \frac{1}{a_m} \sum_{p \in V_m} f_m(p)^2,
\end{aligned}$$

where we can guarantee the inequality for all j since $0 < \lambda_1 < \lambda_2 < \dots$

Note that

$$\sup_m \frac{1}{a_m} \sum_{p \in V_m} f_m(p)^2 < \infty,$$

since $f \in S(K) \subset C(K)$ hence f is bounded and $\#V_m < \infty$ for all m . We can now by dominated convergence, Fubini's theorem, Corollary 2.42 and Lemma 4.2 conclude that

$$\begin{aligned}
\lim_{m \rightarrow \infty} \frac{1}{a_m} \sum_{p \in V_m} f_m(p) (-\Delta_m)^{-2s} f_m(p) &= \frac{1}{\Gamma(2s)} \int_0^\infty t^{2s-1} \lim_{m \rightarrow \infty} \frac{1}{a_m} \sum_{p \in V_m} f_m(p) P_t^m f_m(p) dt \\
&= \frac{1}{\Gamma(2s)} \int_0^\infty t^{2s-1} \int_K f(x) P_t f(x) d\mu(x) dt \\
&= \frac{1}{\Gamma(2s)} \int_K f(x) \int_0^\infty t^{2s-1} P_t f(x) dt d\mu(x) \\
&= \int_K f(x) (-\Delta)^{-2s} f(x) d\mu(x)
\end{aligned}$$

as desired. □

4.2 Main Theorem

With everything in mind we can now prove the convergence in distribution of discrete fractional Gaussian fields to fractional Gaussian fields.

Main Theorem 4.1: Let $s \geq 0$, then X_s^m converges to X_s in distribution in $S'(K)$ when $m \rightarrow \infty$.

Proof. Since $S(K)$ is a nuclear space, by [20, Corollary 2.4] it is enough to prove the convergence of the characteristic functional, e.i that for all $f \in S(K)$

$$\mathbb{E} \left[\exp(itX_s^m(f_m)) \right] \rightarrow \mathbb{E} \left[\exp(itX_s(f)) \right], \quad \text{as } m \rightarrow \infty$$

where $f_m = f|_{V_m}$. It follows from Proposition 3.2 that

$$\mathbb{E} \left[\exp(itX_s^m(f_m)) \right] = \exp \left(-\frac{t^2}{2} \mathbb{E} \left[(X_s^m(f_m))^2 \right] \right) = \exp \left(-\frac{t^2}{2a_m} \sum_{p \in V_m} f_m(p)(-\Delta_m)^{-2s} f_m(p) \right).$$

Likewise, via Definition 3.4 we find

$$\mathbb{E} \left[\exp(itX_s(f)) \right] = \exp \left(-\frac{t^2}{2} \mathbb{E} \left[(X_s(f))^2 \right] \right) = \exp \left(-\frac{t^2}{2} \int_K f(-\Delta)^{-2s} f \, d\mu \right)$$

where we notice by corollary 2.42 that

$$\begin{aligned} \mathbb{E} \left[(X_s(f))^2 \right] &= \int_K (-\Delta)^{-s} f (-\Delta)^{-s} f \, d\mu \\ &= \int_K \int_K f(p) f(q) G_{2s}(p, q) \, d\mu(p) \, d\mu(q) \\ &= \int_K f(-\Delta)^{-2s} f \, d\mu. \end{aligned}$$

The proof now follows from lemma 4.3. □

Chapter 5

Simulations

As a part of the thesis we have created and simulated a discrete fractional Gaussian field on the Vicsek set V_m . The precise field we have constructed was the one given in Example 3.3. These simulations have had a varying rate of success. We have been able to simulate

$$X_s^m(x) = \sum_{j=1}^{N_m} (\lambda_j^m)^{-s} \phi_j^m(x) W_j$$

for a reasonable set of s -values, but we have encountered a limiting factor in the fact that we are not able to simulate X_s^m on the Vicsek set for $m \geq 7$. The reason for this limitation lies in Definition 2.2.

As stated in the definition, for any function $f: V_m \rightarrow \mathbb{R}$ we can define a matrix L_m such that

$$L_m F = \begin{pmatrix} \Delta_m f(x_1) \\ \vdots \\ \Delta_m f(x_{N_m}) \end{pmatrix}, \quad \text{where } F = \begin{pmatrix} f(x_1) \\ \vdots \\ f(x_{N_m}) \end{pmatrix}.$$

It is precisely from this matrix L_m we compute the eigenvalues and eigenfunctions of $-\Delta_m$. Note that the amount of points in V_m is $5^m \cdot 4 + 1$ so as m grows this number explodes, as seen in Table 5.1, and there is no clever way to construct the matrix L_m . Thus the method of constructing the matrix L_m was simply by constructing the Vicsek set V_m and manually checking each point for neighbours.

As we see from Table 5.1, the values explode in size as m grows. The process to check each point is therefore slow. Furthermore the matrix L_m that we construct becomes very large, very fast. Since it needs to hold every permutation of any two points, the matrix L_m becomes a $(5^m \cdot 4 + 1) \times (5^m \cdot 4 + 1)$ matrix. It does however have the property of a sparse matrix, since there are at most five entries in every column, but for $m = 9$ the matrix still fills well over 1 terrabyte in memory.

Having explained the limitations of the simulation, we will quickly delve into the code that generates the simulations. The code will be explained mathematically for clarity, but the full code can be found at:

<https://github.com/Cataclisem/GaussianFieldsThesisCode>.

m	$\#V_m$
1	21
2	101
3	501
4	2501
5	12501
6	62501
7	312501
10	39062501
20	$3.81 \cdot 10^{14}$

Table 5.1: Points in V_m .

5.1 Construction of the Vicsek set for Simulations

The construction is done by first generating the first "plus" $V_0 = \{p_1, p_2, p_3, p_4, p_5\}$ where $p_1 = (0, 0)$, $p_2 = (\xi, 0)$, $p_3 = (0, -\xi)$, $p_4 = (-\xi, 0)$ and $p_5 = (0, \xi)$ for some $\xi > 0$. Then going from V_0 to V_1 we find the endpoints, the tips of the plus, of V_0 and then just add V_0 to it. We can write this formally as:

Let $V_m^{(0)}$ be the endpoints after iteration m and V_m be the points of the Vicsek set after iteration m , we then define a recursive function so

$$V_m = \left\{ y + 2 \cdot x : x \in V_{m-1}, y \in V_{m-1}^{(0)} \right\}, \quad \text{and} \quad V_m^{(0)} = \left\{ 3y : y \in V_{m-1}^{(0)} \right\}.$$

Recall from Definition 3.1 that a discrete fractional Gaussian field is a Gaussian vector indexed by V_m , since the discrete Laplacian depends on the points being in the same places. Thus if we were to compare two different discrete fields we need to make sure the points are in the same order in the list in Python. Luckily we can use the `set` function in Python. Since the `set` function hashes each point, it will put the point in a random place in the set, but it will put the point in the same random spot every time, given that ξ and m are the same each time.

The next step is finding all the neighbours of each point. As stated before, there is no good way to find the neighbours thus this step just iterates over every point. We are therefore going to abuse the symmetry and that we have constructed the Vicsek set outwards¹. This means that all points in V_m , where $x \underset{m}{\sim} y$, are in a distance of exactly $|x - y| = \xi$ from each other. Let $V_{m,p}$ denote the neighbours of a point p . We thus have

$$V_{m,p} = \{x + v \in V_m : x \in V_m, v \in V_0\}.$$

So we actually just add ξ in every direction and see if the point is contained in V_m and finding the neighbours. In Python, we collect each point along with its neighbours in a dictionary.

¹In Section 1.2 when we theoretically constructed the Vicsek set, it was constructed inwards. But when programming the Vicsek set, it was easier and faster to construct outwards.

This is because it provides us with a convenient structure to work with, but also because it allows for efficient lookups since dictionaries in Python are hash maps. So while we add each point and its neighbours to the dictionary, we give each point a name, which is x_i where i denotes its place in the set. So each dictionary entry has the following information:

$$\left\{ \text{name: } x_i, \text{ Neighbours: } \{(x_\alpha, y_\alpha), \dots\}, \text{ coordinate: } (x_i, y_i) \right\}.$$

To actually construct the discrete Laplacian matrix we do a `list comprehension` with parameters exactly as in Definition 2.2.

With the construction of the Vicsek set, the neighbours of each point and the discrete Laplacian, we can move on to generating the eigenfunctions and eigenvalues.

All the constructions mentioned above can be found in `code/VicsekSet/VicsekSet.py` and `code/simulations.py`.

5.1.1 Eigenvalues and Eigenfunctions

The way we find eigenvalues is through the `numpy.linalg.eigh` function. The function is for finding eigenvalues and eigenvectors of real symmetric matrices. Some of the best eigenvalue finding algorithms have a runtime of $O(n^2)$, where n is the number of rows in the matrix. As $5^m \cdot 4 + 1$ is a number that grows fast as m increases $(5^m \cdot 4 + 1)^2$ grows even faster making the algorithm slow as m increases.

The code we are describing can be found in `simulations.py` line 95-101 and line 114-133. Now the function `numpy.linalg.eigh` gives us a vector of eigenvalues v_{eig} and a matrix where each column is an eigenvector M_{eig} , having sizes $(5^m \cdot 4 + 1) \times (1)$ and $(5^m \cdot 4 + 1) \times (5^m \cdot 4 + 1)$ respectively. The matrix of eigenvectors is actually also our eigenfunctions, since

$$-\Delta_m(M_{\text{eig}})_{i,-} = (v_{\text{eig}})_i \cdot (M_{\text{eig}})_{i,-}, \quad \text{so } -\Delta_m(M_{\text{eig}})_{i,j} = (v_{\text{eig}})_i \cdot (M_{\text{eig}})_{i,j} = \phi_i(x_j).$$

Thus we have both our eigenvalues and eigenfunctions.

Calculating the discrete field in each point is the exact same as in Example 3.3, where the white noise W_i is generated by `numpy.random.standard_normal`. To generate the heat maps we plot our points with `matplotlib` and lay in a colorbar where the colour value corresponds to the values from $X_s^m(x)$ for $x \in V_m$.

5.2 Simulations on the Vicsek set

The point of these simulations is of course to show the convergence of X_s^m to $\widetilde{X}_s$ in distribution, but as we have not been able to simulate much over $m \geq 7$ we don't really see any convergence in distribution. What we can observe instead is that the variance in the fields increases as s increases.

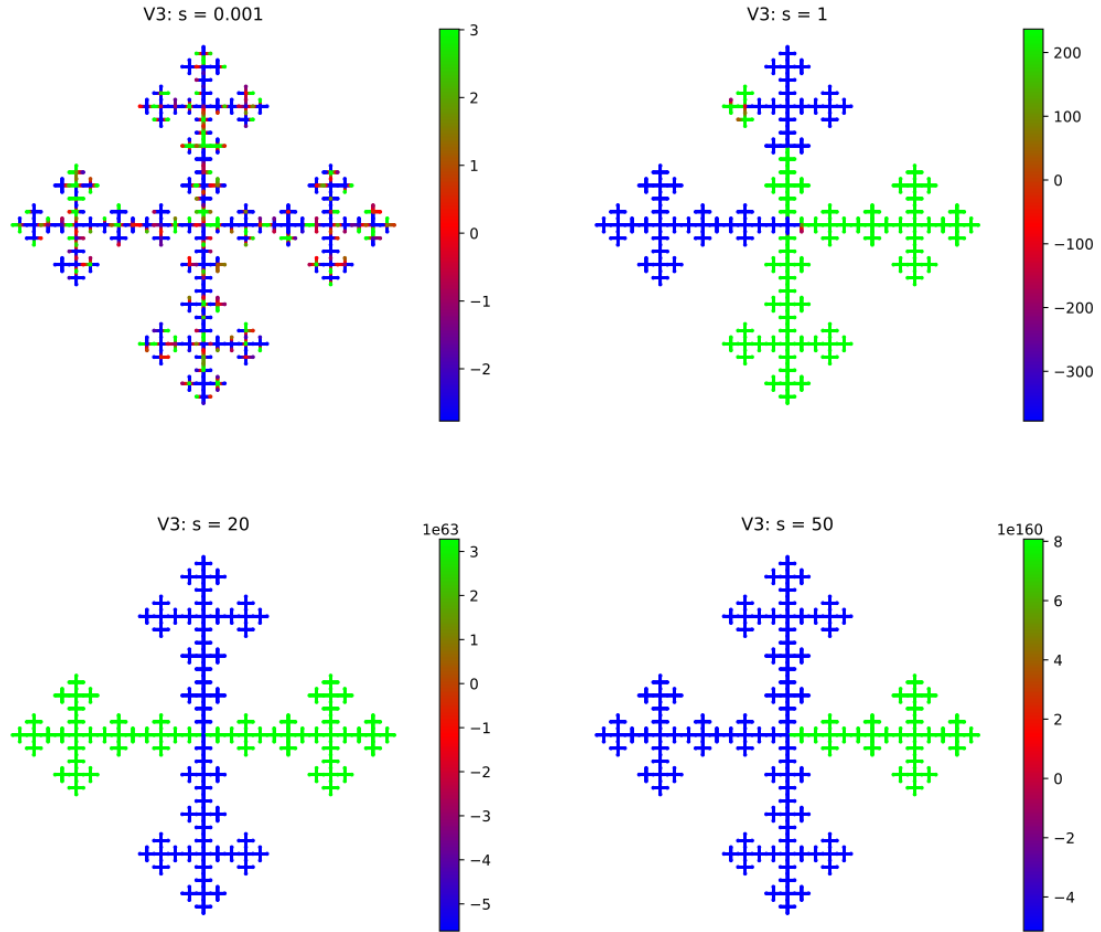


Figure 5.1: Simulations of the Vicsek set on V_3 with different s parameters and 4 different white noise sequences.

Figure 5.1 shows 4 different Vicsek sets on V_3 each simulated with their own white noise. As can be seen from the values on the colorbar, the fields increases in variance as s increases. While this is not difficult to see from the definition of X_s^m , what is interesting is the variance when s is small. If we look at $s = 0.001$, we find that there are many more red spots, meaning there are more transitions between values.

The figure with $s = 50$ is very much just green and blue. So the simulations have their extreme values, but looking at the colorbar we discover that the reason that they are so

extreme is because the difference in their values is about 10^{161} . Hence the field has grown very large in each direction. This again makes sense looking at X_s^m where the value $(\lambda_j^m)^{-s}$ dominates each term as s grows larger.

We can further study the influence of the white noise on the simulation. In Figure 5.2 and Figure 5.3 we have simulated X_s^m on V_3 for two different sequences of white noise and 4 different values of s . If we consider $s = 0.001$, in both Figure 5.2 and Figure 5.3, we see that the values are quite spread out on the Vicsek fractal. At the value $s = 1$, both figures have their values quite far apart, with values being around 400 apart. But even as they have the extreme values, they are each contained within the same area and do not leave it. For $s = 20$ we see that both figures start to diverge at the same rate, which makes sense as we argued in the case with Figure 5.1.

A peculiar thing happens as we go from $s = 1$ to $s = 20$. It is that the two blue crosses on the left of the Vicsek set in Figure 5.2 and the small green section on the right of the Vicsek set on Figure 5.3 both turn into their opposite colour at $s = 20$. This is an effect we cannot quite explain, but we suspect that it might have to do with *floating point errors* as when $(\lambda_j^m)^{-s}$ becomes very large then $(\lambda_j^m)^s$ becomes very small.

But again the simulations act as we expect. This is because $(\lambda_j^m)^{-s}$ only changes as s changes, and the eigenfunction $\phi_i(x_j)$, with $x_j \in V_m$ is the same each time. Thus the only change in the fractals is when w_i changes. But since w_i is constant, it follows from the same argument as with Figure 5.1 that $(\lambda_j^m)^{-s}$ becomes the dominant factor as s changes.

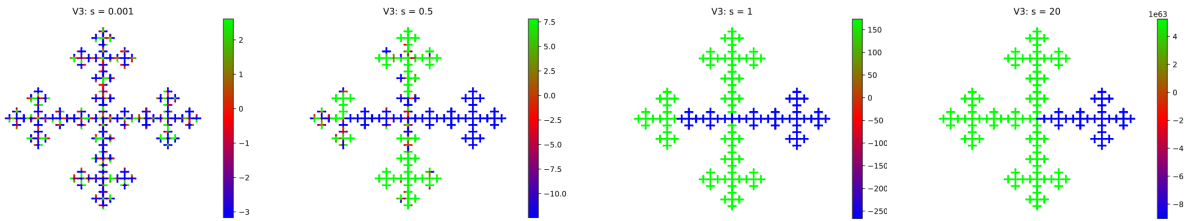


Figure 5.2: Simulations of the Vicsek set on V_3 with different s parameters and with the same white noise. A larger version of the image can be found in Appendix A.

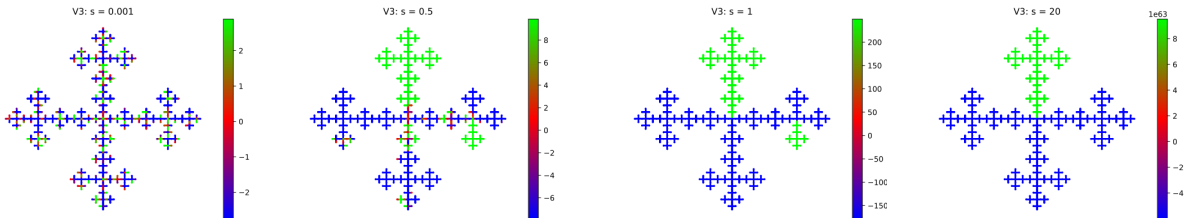


Figure 5.3: Simulations of the Vicsek set on V_3 with different s parameters and with the same white noise. A larger version of the image can be found in Appendix A.

5.2.1 Simulations on the Sierpinski gasket

The described theory in this thesis, is as mentioned before, an extension of the paper [23] by Fabrice Baudoin and Li Chen. Therefore the theory that made the simulations of the Vicsek set possible also hold for the Sierpinski gasket. Again, we are not able to simulate m very large, but since the amount of points on the Sierpinski gasket is "only" $\frac{3(3^m+1)}{2}$, we are able to simulate $m \leq 11$. The code for simulations of the Sierpinski gasket is almost identical to the code for the Vicsek set. The main difference being the way the fractal is constructed and the way neighbours are found.

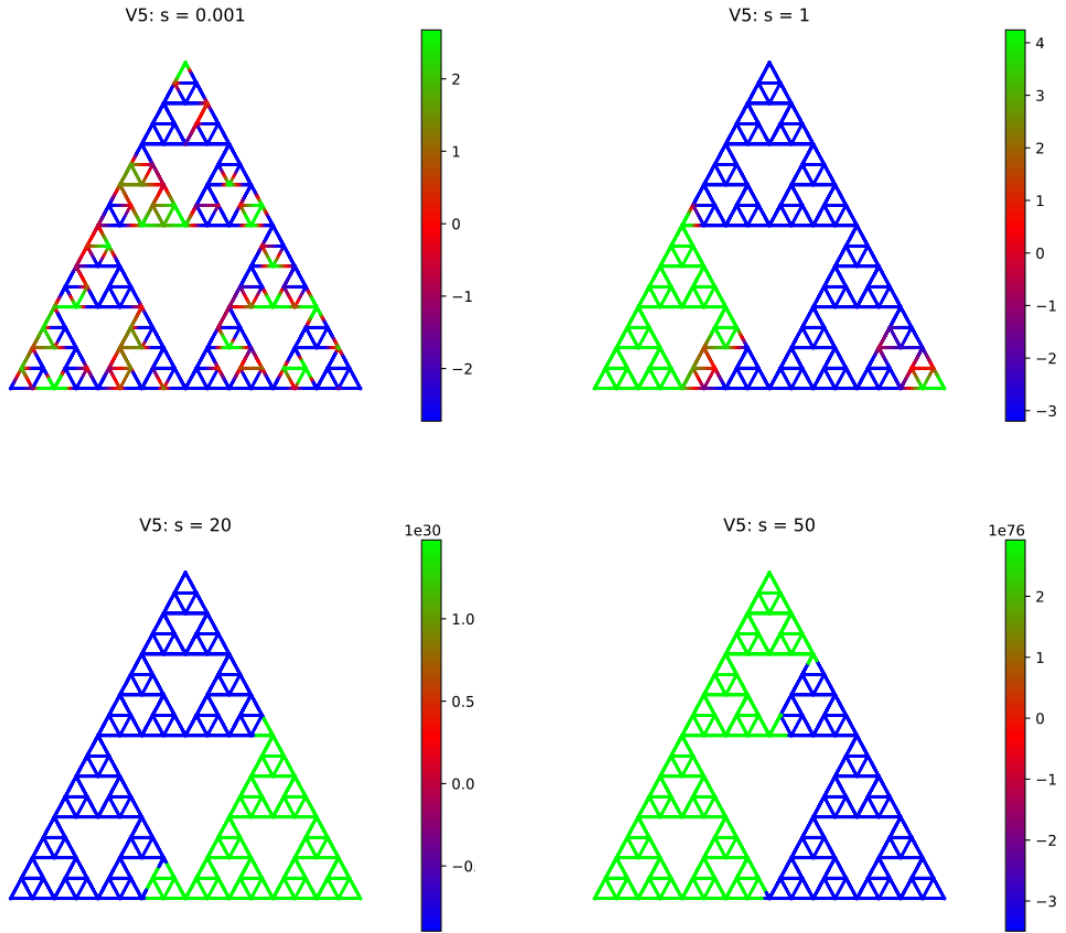


Figure 5.4: Simulations of the Sierpinski gasket on V_5 with different s parameters and 4 different white noise sequences. A larger version of the image can be found in Appendix A.

Again, we see in Figure 5.4 that the variance diverges as s increases, which as explained in the simulations of the Vicsek set, was to be expected. They start with a large variance at $s = 0$ and as s increase the variances begins to increase.

If we study the influence of white noise in the simulation of the Sierpinski gasket, we see in Figure 5.5 and Figure 5.6, without surprise, that it behaves almost exactly the same way

as the Vicsek set. Figure 5.5 and Figure 5.6 show some variance when $s = 0.001$ but end up with clusters of values in the end.

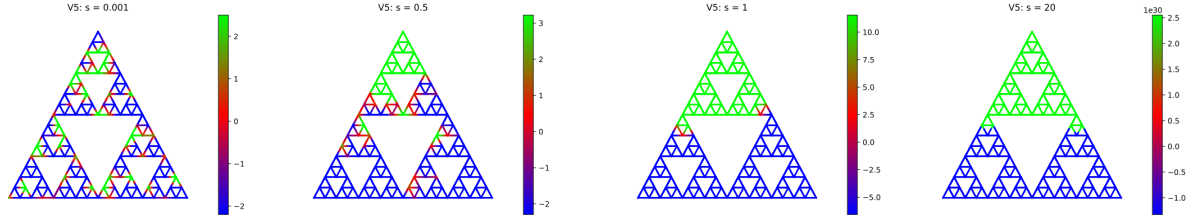


Figure 5.5: Simulations of the Sierpinski gasket on V_5 with different s parameters and with the same white noise. A larger version of the image can be found in Appendix A.

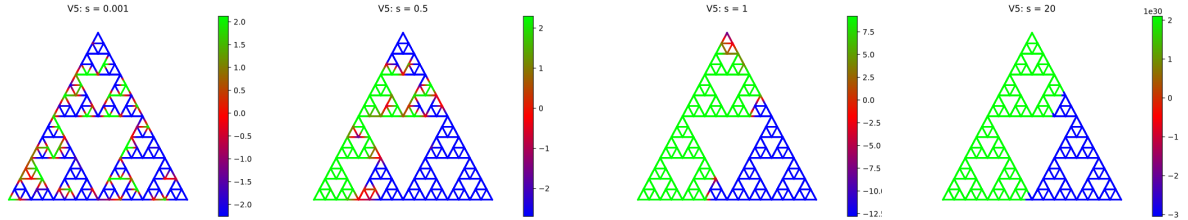


Figure 5.6: Simulations of the Sierpinski gasket on V_5 with different s parameters and with the same white noise. A larger version of the image can be found in Appendix A.

5.3 Conclusion and Future work on the Simulations

While it was unfortunate that we were not able to simulate convergence of distribution we were still able to find some points of interest in the simulations. The first being how quickly the simulations diverge and clusters from quite a varied state. Further, one can see a clear influence of the white noise when observing what values the simulations converge to.

Lastly, the simulations on the Vicsek set and the Sierpinski gasket ended up coinciding. This is of course what we would expect as the theory behind the simulations are the same. But the construction of the Vicsek set and the Sierpinski gasket could have been different enough to cause some inconsistencies between the two simulations.

Should one wish to continue the work on the simulations, there are three questions to tackle.

The first is the clustering, and why every simulation ends with the clusters of opposite value. The question is probably a more mathematical question rather than one that can be answered through computer science. Since the code should not have any influence on the way that the white noise is chosen, the clustering might come from a property of the eigenfunctions or eigenvalues. It is of course also possible that these clusters are somehow caused by floating point errors as the some of values used in the simulations become very small.

The second question to explore is why a point, encapsulated by points of opposite values, suddenly become that value when the simulation progresses. This can be seen in Figure 5.2 and Figure 5.3 where we see areas that become engulfed by the opposite one. This question relates itself very much to the first question, and one might suspect that finding a solution to the first question will also answer this one.

Lastly one could extend the simulations to other fractals. Much of the theory derived in the thesis is not specific for the Vicsek set and could be expanded upon to cover other self-similar fractals.

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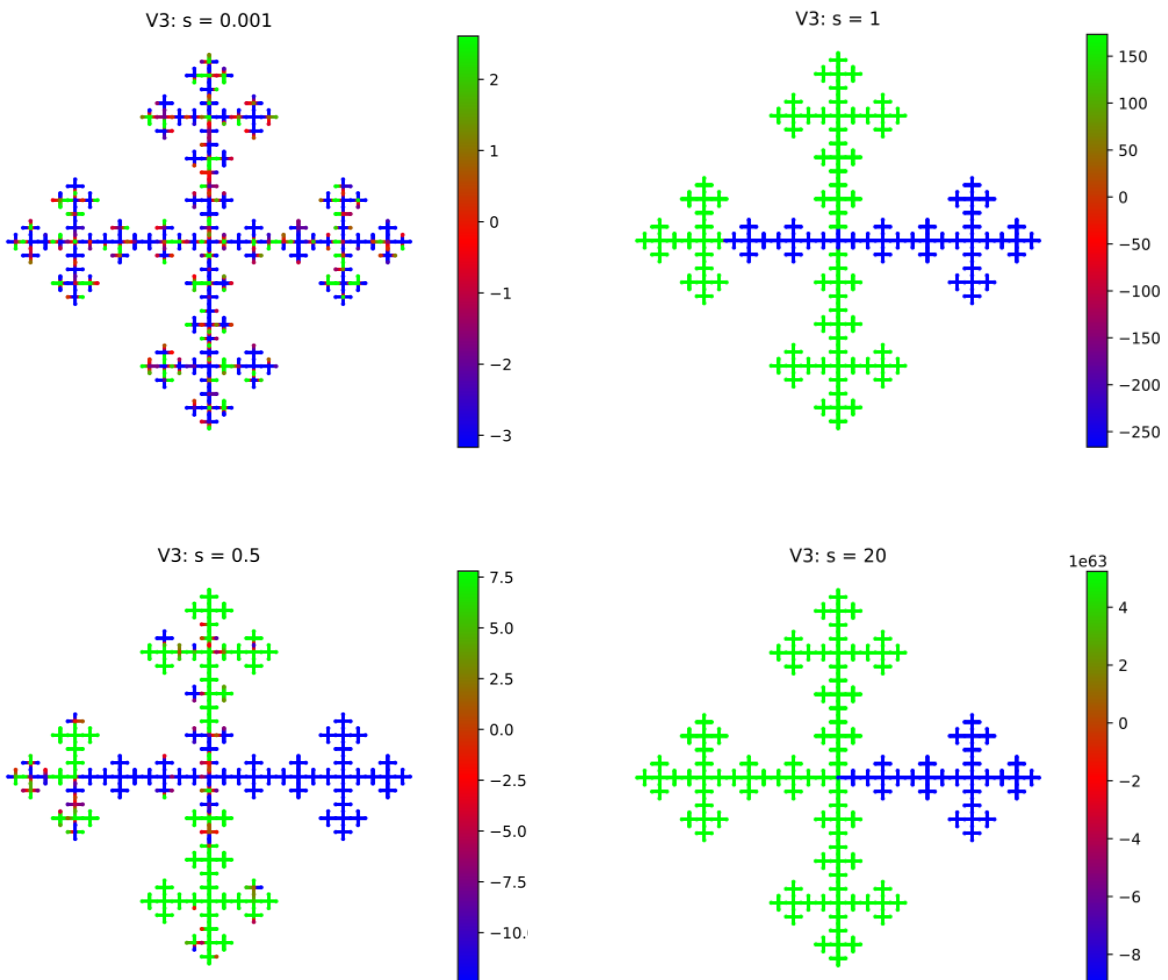
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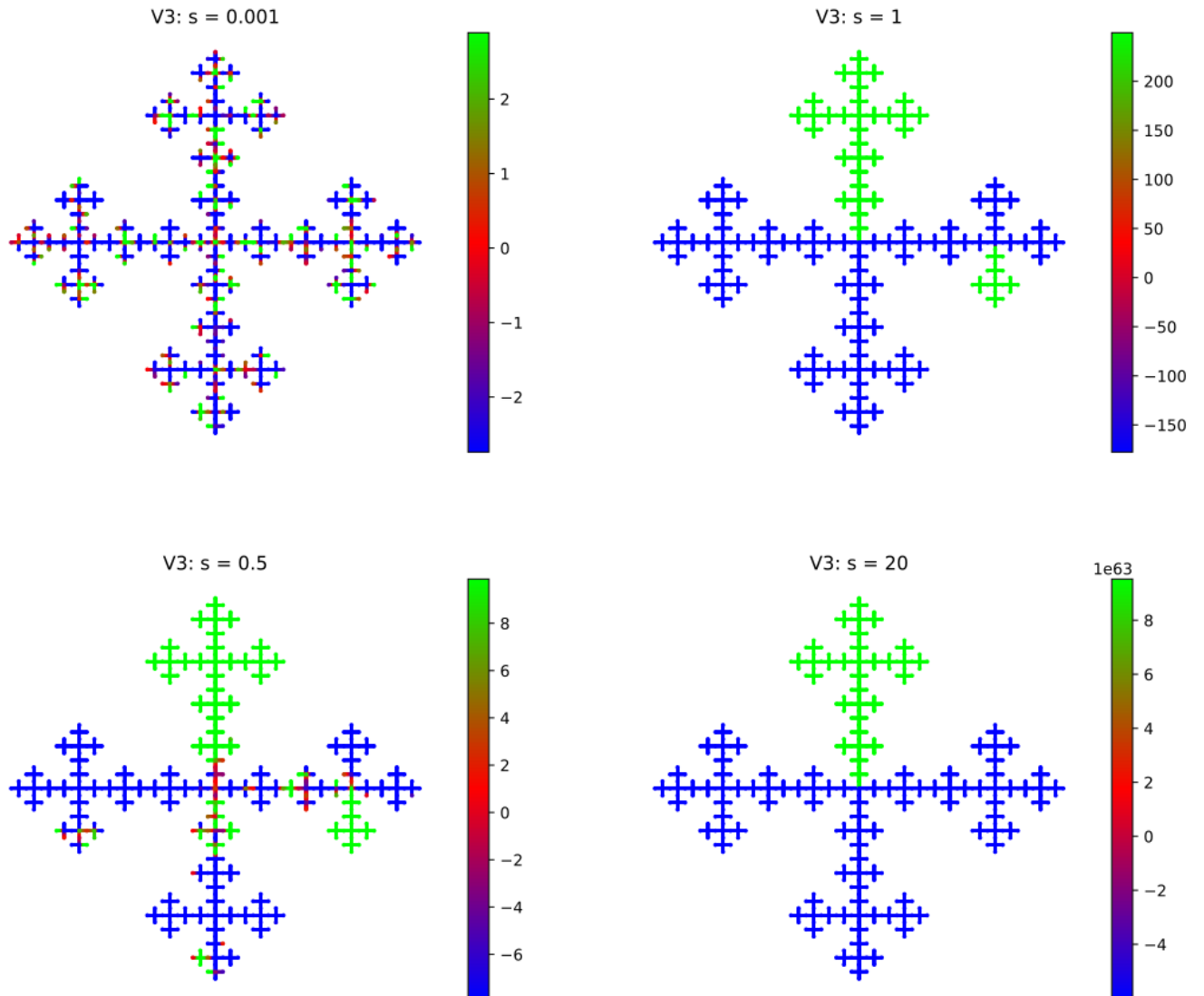
Appendix A

In this appendix we have included larger version of the images from Chapter 5.

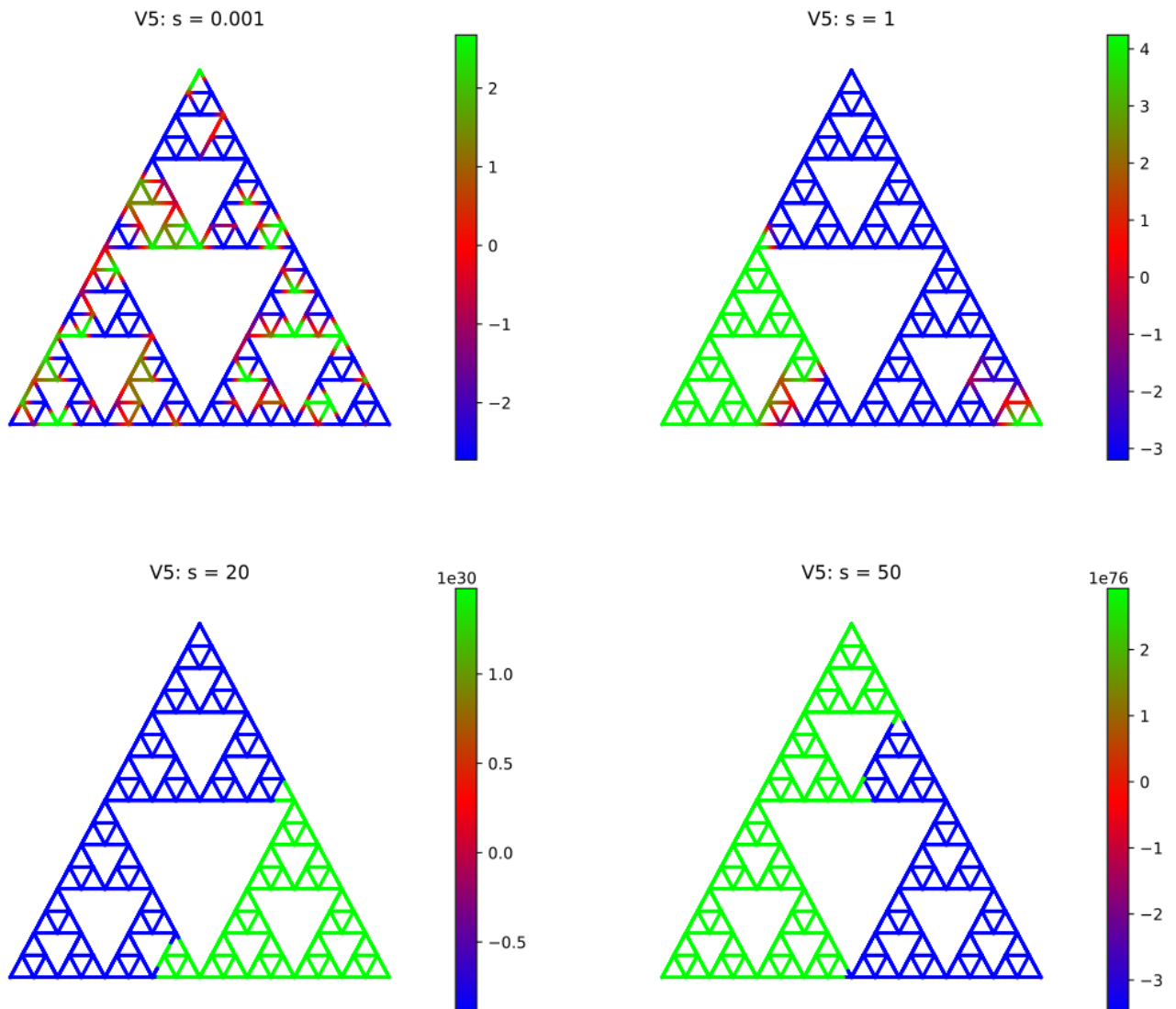
Full scale image of Figure 5.2



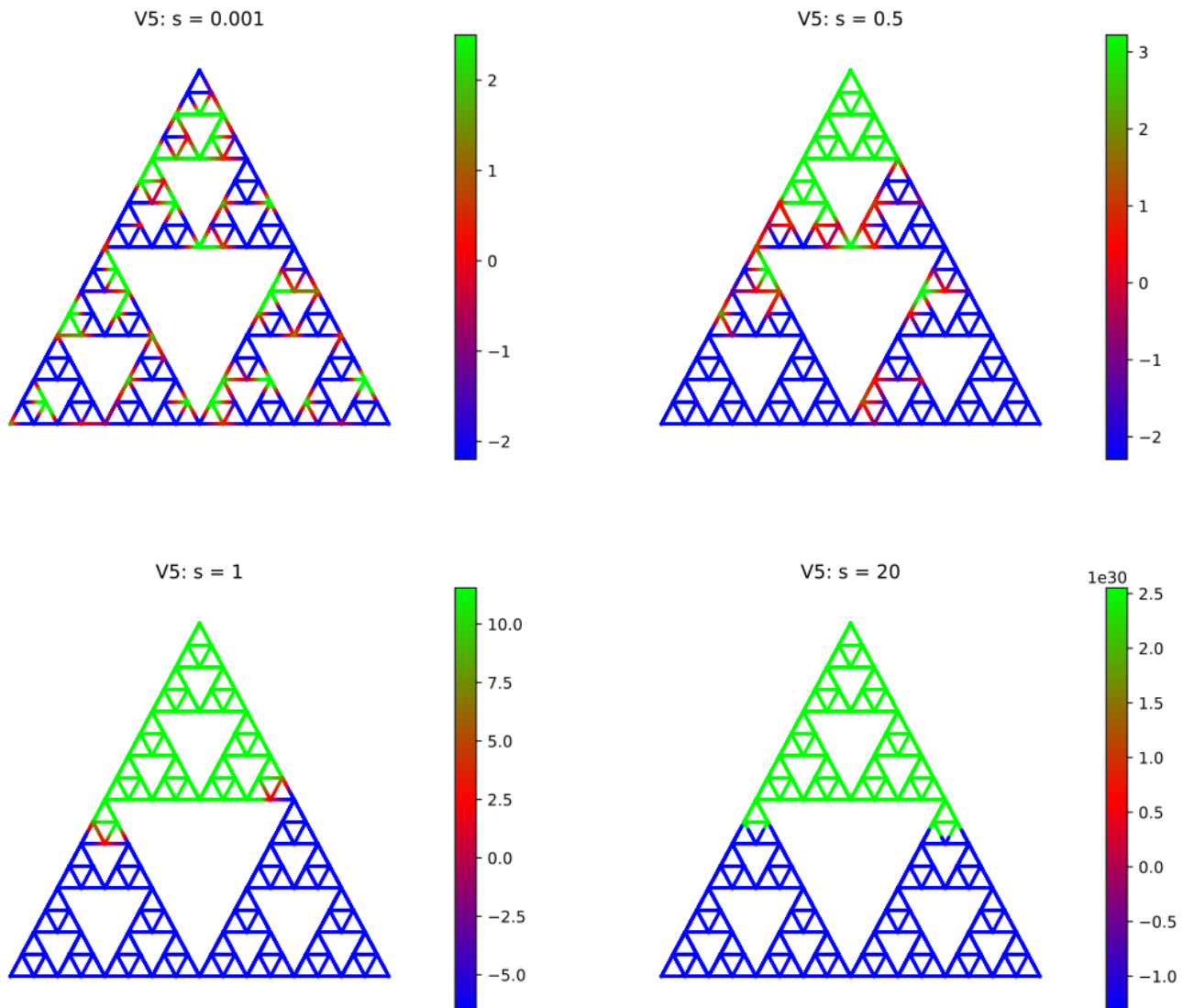
Full scale image of Figure 5.3



Full scale image of Figure 5.4



Full scale image of Figure 5.5



Full scale image of Figure 5.6

